

MRC National Survey of Health and Development (MRC NSHD)

Dietary data report 2006-11

Prepared by

Celia Prynne, Wing Nip, Alison Lennox

Cambridge

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1 Executive summary

1.1 Introduction

Assessment of dietary intake of cohort members has been an important component of the MRC National Survey of Health and Development (NSHD), particularly in adulthood. The clinic visit or having a home visit conducted in 2006-11 has enabled a fifth assessment of diet, the earlier timepoints being at age 4 years in 1950, at age 36 years in 1982, 43 years in 1989 and 53 years in 1999. These repeated measures of diet enable trends in intakes to be examined and relationships to health measures to be explored, of particular interest now that the cohort is reaching its ageing years.

1.2 Aim of report

This report outlines the procedures used to collect, code and analyse dietary information from the assessment of NSHD cohort members at age 60-64 years. It describes the consumption of foods and food groups and intake of nutrients for the 2006-11 timepoint and relates the intakes of macronutrients to dietary goals and of micronutrients to Dietary Reference Values (DRVs). It also compares the intakes in 2006-11 to the survey carried out in 1999 to provide indications of how the diets of the cohort members have changed over the intervening 7-10 year period.

1.3 Methods

Dietary intakes were collected as 5 day estimated (unweighed) diaries, as they were in the previous adult assessments, including two weekend days if possible. Diaries were given to cohort members at the end of the clinic visit, with nurses giving detailed instructions on completion of the diary and the level of detail required. Cohort members sent the completed diaries directly to HNR in prepaid envelopes; these were logged on arrival. Diaries were coded and edited by HNR's dietary assessment team, using DINO (Data In Nutrients Out), HNR's purpose built Access based dietary assessment system, incorporating food composition tables from McCance and Widdowson's The Composition of Foods 6th edition. Resulting food and nutrient intakes were averaged for the days of collection and reported as mean intakes for 5 days. Results were compared, where possible, to those from the 1999 assessment, statistical analysis being carried by ANOVA.

1.4 Results

1869 cohort members (880M, 989F) completed at least 3 days of diet record, 84% of those attending the clinic visit. Of these 1318 (70.5%) (594M, 724F) also completed at least 3 days in 1999.

For those foods that can be compared to 1999 (some food groupings were different between the two timepoints), consumption of white bread was lower and granary and wholemeal higher in 2006-11 than in 1999, and there was a large reduction in consumption of cakes and sweet puddings. Milk consumption decreased and was predominantly semi-skimmed; cheese consumption was unchanged but yoghurts increased. Consumption of both red and white meats was reduced from 1999, but there was no change in fish. Vegetable consumption was higher, but potatoes lower. Coffee consumption was lower.

In terms of energy and nutrients, energy intakes were somewhat lower in 2006-11 compared to 1999 and represented 88-89% of the Average Energy Requirement (AER) for this age group for the UK. Energy intake has been found to be less than energy expenditure as measured by doubly-labelled water when assessed using dietary records, as it is with other detailed methods of dietary assessment, such as 24 hour recall, but here is no indication that the degree of underestimation has changed from previous assessment of diet in the cohort¹. Intakes of total fat were little changed since 1999, but saturated fat intakes were reduced, with a lower percentage of individuals above the recommended 10% of energy in 2006-11 than in 1999. Intakes of non-starch polysaccharide (NSP) (dietary fibre) were higher than in 1999. There was no difference in alcohol consumption, although the types of drinks consumed were slightly different, with increases seen for wine. Intakes of vitamins were higher in some cases and overall the percentages of cohort members with intakes below the Lower Reference Nutrient Intake (LRNI) values were lower than in 1999. For minerals, there were few changes, and the percentage of individuals with intakes of potassium and zinc below the LRNI was increased compared to 1999.

1.5 Conclusions

A successful assessment of dietary intake of the NSHD cohort members was carried out as part of the clinic visit at age 60-64 years from 2006-11, with good responses rates to the request to complete the diet diary and well completed diaries, which allow prompt and complete coding and editing, and providing a complete dietary dataset by the Summer of 2011. Dietary data indicate that food and nutrient intakes are largely similar to those of 1999, with small decreases in energy intakes, which would be expected for this age group compared to earlier adulthood. Trends from 1999 indicate that dietary

¹ A Stephen, B Teucher, L Bluck, D Cole, E Fitt, A Mander, R Woodward, A Wright, B Bates, C Roberts, H Mackenzie, C Deverill, J Mindell . National Diet and Nutrition Survey Rolling Programme.Comparison Study: A comparison of results by dietary assessment method: repeat 24-hour recall and four-day estimated diet diary. Report prepared for the Food Standards Agency. 2008

intakes in 2006-11 are somewhat healthier than the earlier timepoint, with lower saturated fat and higher fibre intakes, and a lower proportion of individuals with low intakes of vitamins. The dietary data for NSHD now provide opportunities for examining the associations between lifetime diet and health outcomes, particularly in relation to early old age.

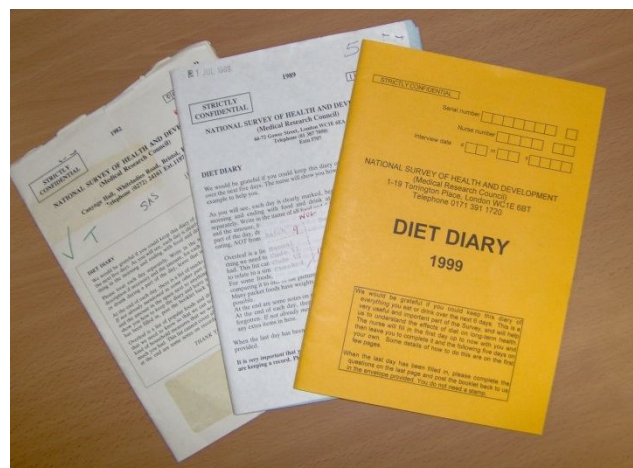
2 Introduction

2.1 Background

The MRC National Survey of Health and Development (NSHD), or the 1946 British birth cohort is the longest running birth cohort in the world, with data collected on numerous occasions since the birth of cohort members in one week of March 1946.

Dietary data had been collected at ages 4, 36, 43 and 53 years. At age 4 years, a brief 24 hour dietary recall was conducted to mothers of the cohort members. In adulthood, an estimated (unweighed) diet diary was used. At ages 36 and 43 years, this consisted of 2 days of recall and 5 days of prospective recording; at age 53 years only the 5 days of prospective recording was collected, and in general the 5 days of prospective recording have been used in publications of dietary data from all adult timepoints much more frequently than the 2 days of recall.

The first adult assessment in 1982 was coordinated and administered by NSHD in Bristol. A 5 day estimated (unweighed) diary was used with an interviewer assisted recall of the 2 days immediately preceding the interview. The diary, which asked for information on portion sizes to be given in household measures, was adapted from the weighed food assessments used in other studies. The dietary information was converted into food codes and weights by manual coding before being converted into electronic form for calculation of nutrient intakes. In the mid-1980s, the Dunn Nutrition Unit in Cambridge became more involved with the dietary component of NSHD and at the next assessment in 1989, the diary was modified to include some photographs to assist portion size estimations. A computer program, DIDO (Diet In and Data Out), designed in Cambridge, was used to enter food data directly; the output files being compatible with the Dunn's in-house nutrient analysis program, MW4. All coding was carried out at the Dunn, under the direction of Alison Paul. In 1999, the Dunn again carried out the dietary coding and analysis aspects of the data collection. The diary was very similar to that used in 1989, although there were a few differences such as the 1999 diary having a box specifically for recording supplement intake, which was not available



The diet diaries used in NSHD

in the 1989 diary. At the same time the output file from the previous coding of the 1982 diaries was converted into a form compatible with the later DIDO output files. This

enabled the 1982 data to be re-analysed in parallel with the 1989 and 1999 data using the MW4 program.

In the mid-1990s, funding became available to return to the 24 hour recalls conducted in 1950 and the questionnaires which included these recalls were retrieved. Dr Celia Prynne was recruited to work with Alison Paul to code the recalls and prepare and write up the results of the 1950 assessment, when the cohort members were 4 years old. This was completed in the late 1990s, just before the 1999 assessment. In 1998 MRC Human Nutrition Research (HNR) was created to undertake human nutrition research in Cambridge; included under HNR's umbrella was dietary assessment using DIDO and projects such as NSHD. Hence the MRC in Cambridge, first as the Dunn Nutrition Unit and then as HNR has extensive experience with the dietary intakes of the cohort and has been involved with all publications involving dietary data since the mid-1980s.

As cohort members moved into their ageing years, it was decided to conduct an in-depth assessment of physiological and psychological functioning of the cohort in order to provide insight into the normal ageing process. Nutrition over the life course plays an important role in determining the outcomes of ageing, particularly through its role in chronic disease development. As the NSHD collaborator with the experience of and responsibility for the nutritional aspects of the cohort, HNR undertook the planning and coordination of dietary data collection as part of the clinic visit which took place between 2006 and 2011, as well as the coding and analysis of dietary data.

A 5-day diet diary, similar to that used in previous assessments of the cohort was used to record dietary intakes. With the increased recognition of how eating environment may influence the quantity and quality of food consumed and hence nutrient intake, extra information was collected about characteristics associated with each eating occasion.

2.2 HNR responsibilities

The collaboration between HNR and the MRC Unit for Lifelong Health and Ageing (LHA), which now hosts NSHD, is laid out in a Memorandum of Understanding (MOU) between the two units. This details HNR's responsibilities and obligations for the dietary data collection for NSHD, including that at age 60-64 years:

- Providing specialist nutrition and dietary advice, analysis, interpretation, and oversight as appropriate, on LHA scientific programmes through the Diet Project Management Group

- Initiating and developing research proposals and to provide specialist nutrition and dietary collaboration on research proposals from external researchers through the Diet Project Management Group
- Contributing to governance through membership of the Diet Project Management Group, as outlined in the NSHD draft guidance on access to NSHD data and biological samples
- Encouraging collaboration with other external researchers working on NSHD dietary data and be the point of contact for other researchers who may wish to undertake collaborative research projects within LHA programme 6
- Using HNR's expertise in nutrition, and dietary data collection, to ensure credibility of the dietary data in NSHD and that the data are used and interpreted appropriately in all aspects relating to human nutrition and health, through HNR's involvement with the programmes and projects and membership of the Diet Project Management Group
- Providing scientific advice and guidance on research governance as appropriate, taking responsibility for ensuring that all legal and regulatory requirements are met for specific programmes and projects as agreed with LHA, and particularly where HNR is the lead investigator
- Creating and delivering the NSHD dietary dataset at 60-64 years to LHA in a mutually agreed format within agreed timeframes of releasing an interim and a full dataset when 50% and 100% data were received respectively at HNR

The agreed timeframes of delivering the NSHD dietary dataset at 60-64 years to LHA at 50% (interim dataset) and 100% (full dataset) target points were revised in December 2010, where the final number attending clinic and home visits was reassessed to a fewer number than originally projected and diet diary coding proceeded faster than expected. An interim dataset would therefore represent more than 75% of the total survey data rather than 50% by the time fieldwork was completed. In the view of this, it was agreed between LHA and HNR that, instead of creating both interim and full datasets, only a single full dietary dataset would be produced and released at an earlier timepoint (end of September 2011).

3 Methodology

3.1 Background

Dietary data were collected as part of the clinic visit for the NSHD assessment at age 60-64 years. Consistent with previous method of assessment, dietary intakes were evaluated using a 5-day estimated (unweighed) diet diary. The diet diary used for this data collection was the current HNR diary; this had been modified from the previous assessments of the cohort to yield information on where foods were eaten and who was present during the eating occasion.

3.2 Placing the diet diary

The diet diary was placed towards the end of the clinic or home visit. Prior to the appointment, it was determined whether or not each cohort member had completed a diary in 1999, as this would affect the level of discussion and detail needed for completion of the diary when it was placed. Cohort members were asked to record 5 days of dietary intake for the next 5 days, i.e. beginning the day after the clinic visit, or if not the next day, at least within the next week. Each diary was provided in a zip lock plastic wallet for the collection of package labels. This also included the stamped envelope for the return of the completed diary to HNR. A flyer explaining the reasons why dietary information was collected was also given to cohort members to encourage participation (Appendix A). Each diet diary and zip-lock plastic wallet was labelled with the HNR number, which was created at HNR to enable identification of NHS numbers and cohort members' names for correspondence regarding blood samples. Since dietary data throughout the survey are identified by NSHD serial number, the HNR number was later converted to serial number for each diary. The appointment date was also recorded inside the front cover of the diary as a second measure of identification.

The nurses who handed out the diaries had been trained in the method of diet recording and explained the procedure to the cohort members. A completed example of a day's food record was provided at the beginning of the diary. Each day was separated into seven time slots for entry and cohort members were also asked to record the exact time of each eating occasion. The nurses demonstrated the various ways that portion size could be estimated:

- in household measures (e.g. one tablespoon of baked beans, one Kit Kat (two fingers))
- for packaged foods to note the weight indicated on the packet
- photographs provided in the diary for 15 frequently consumed foods as small, medium and large portion sizes which cohort members could use for identical or similar foods

For each eating occasion, cohort members were asked to record where they were and who else was present at the time of eating.

At the end of each day there was space to record ingredients for recipes of dishes made at home by the cohort member and also space for any dietary supplements taken that day (Appendix C). Attention was also drawn to the checklist at the end of the diary where cohort members were asked to record their usual eating habits (for example, type of milk or fat spread usually consumed) to facilitate coding in cases where details were omitted in the diet record. They were asked to record brand names for foods wherever possible and were asked to collect the food label information/wrappers for any unusual foods and ready meals consumed to help coders identify or clarify items.

Instructions for clinic nurses when discussing the diet diary with cohort members were provided in the nurse manual, as given below:

DIET DIARY INSTRUCTIONS

Introduction: Explain that we would like them to keep a diary for 5 days over the following week so that we can get a true picture of what the cohort member really eats. It is also important to get as much detail as possible. This means getting:

- information not only about food and drinks taken at meals, but also about anything eaten or drunk between meals
- details about food and drink; for example, if a Chinese take-away was eaten we need to know what things they ate (e.g. sweet & sour pork, prawn crackers); or if someone's had a burger it's very helpful to know who made it (eg Macdonald's, Burger King, home-made)

Encourage the participant to keep the diary about food and drink for 5 days over the next week (ideally 3 weekdays and the 2 weekend days). If they are uncertain, then leave the diary anyway, and encourage them to fill in as many days as they can, even if not all 5. Establish whether they will be away on holiday. If they are, ask them to fill it in when they return. If working away from home is usual, then ask them to fill in the diary at their workplace. If they agree then:

Protocol:

- Place the label for the diary FD N and the interview date inside the front cover of the diary

- Ask them NOT to write anything on the front cover of the diary
- Explain that they should fill it in each day, if possible as they go along during the day, and give the kind of level of detail that the example used
- Show them the examples and the pictures of portion sizes in the front of the diary
- Show them the general questions on the last page of the diary. This is important as they may not see these questions and they help with coding of the diet record
- Emphasise the importance of not changing what they usually eat just because they are keeping a diary
- Place the label FDL N on the zip folder and give it to them for placing the labels for foods they consumed during the week. This is especially important for any ready meals
- Show them the flyer which explains why we need the diet information and encourage them to complete and return the diary
- Instruct them about the return of the diary. The diary and the zip folder should be placed in the stamped, addressed envelope inside the zip folder and posted back to HNR

3.3 Tracking the diet diary

All cohort members, seen at the clinics or at home, were asked to complete a diet diary. Initially, nurses were asked to notify HNR whenever a diet diary was taken by a cohort member via the tracking documents that were faxed to HNR. However, this proved to be challenging and led to a change in protocol in November 2010 where HNR would only be notified if the cohort member declined to participate in the dietary assessment. Nurses were instructed to record this information explicitly on either the fax cover sheet or the Blood Sample Tracking form, where information was entered into the NSHD database along with other tracking data by the HNR data entry team.

The completed diet diary was returned to HNR by post in a pre-paid envelope given to the cohort member during their clinic or home visit. Once received, HNR personnel logged the diet diary firstly into the NSHD database using the HNR number attached to the diary; date of receipt, the numbers of diary days completed, the numbers of food packages returned and whether or not the general questions on the last pages of the diary were completed, were then entered to this database. This diary was subsequently relabelled with the corresponding NSHD serial number, which is a unique number given to the cohort member at birth, when they were enrolled into the study. The task of converting the numbers on the diet diaries from the HNR number to the serial number was originally carried out by LHA staff; this was subsequently taken over by HNR personnel. The relabelled diaries were logged onto the HNR in-house dietary assessment

system DINO (Diet-In-Nutrients-Out) where each step of data entry was tracked (i.e. coding and editing).

3.4 Dietary data processing

Once the received diaries were logged at HNR, they were passed to HNR's dietary assessment team to be coded by trained coders and editors. Food intakes were entered into a modified version of HNR's dietary assessment system DINO (Diet In Nutrients Out), an all-in-one dietary recording and analysis system written in Microsoft Access. DINO is an updated system from the DIDO system used previously; it is based on the same principles as DIDO, but is more versatile and user friendly. The food composition data used was based on McCance and Widdowson's "The Composition of Foods", 6th edition² with additions where necessary for new foods (see below).

Coders attempted to match each food or drink item recorded in the diary with a food code and a portion code from DINO. For composite items, which could be split into their component parts, for example sandwiches, each individual component was assigned. If an item had been recorded and there was no suitable code in DINO or there was insufficient detail to code the food, the entry was flagged as a query.

Within DINO, each food code is linked to appropriate portion size descriptors, mainly household measures, which are then linked to the correct weight for that portion or, where the portion size is described as a weight, the weight can be entered directly in grams. The coding of portions described as small, medium or large was based on the Food Standards Agency's reference book "Food Portion Sizes"³. Weights for common branded foods from supermarkets and manufacturers and for foods from fast food outlets were obtained from the relevant website or if necessary, a food would be purchased and weighed.

Where the coder could not resolve the food or portion consumed, the entry was flagged as a query for action by an editor who had greater nutrition background and experience. The editors assigned appropriate codes for all flagged food and portion codes and checked any other queries raised by the coders. In general, where details for the coding of foods were missing, formally agreed default codes were used, such as for the type of milk in tea or coffee in a café, or lasagne eaten at a restaurant. Where portion sizes were missing, an estimate was made using the same weight if the food was consumed

² Food Standards Agency (2002) *McCance and Widdowson's The Composition of Foods. Sixth Summary Edition*. The Royal Society of Chemistry, Cambridge.

³ Food standards Agency (2002) *Food Portion Sizes by Ministry of Agriculture, Fisheries and Food. Third Edition*. London: The Stationery Office (TSO).

by that cohort member on another dietary day, or a portion size consistent with the cohort member's usual consumption (e.g. small, medium or large) or an age-appropriate average portion.

For a new product not in DINO, editors visited supermarkets or contacted the manufacturer to obtain information on the nutrient content in order to decide whether a new food code was needed. This decision was based on the nutritional composition compared to that of existing codes and the frequency of consumption. If a new food code was required, the nutrient content was entered into the databank. If a portion was used but there was no corresponding portion code on DINO, a new portion code was created using either a weight from an equivalent food, or the food item was weighed and the weight entered into DINO for future use.

Where a cohort member consumed a homemade recipe, each individual food ingredient was linked with the food group of the recipe. DINO has the capability to report on these foods both at the recipe level and food level, providing a clearer picture of consumption of components like meat, fish, fruit and vegetables.

For a homemade dish where a recipe had been recorded, the ingredients were entered individually using the appropriate cooked food codes, and all the codes for the dish were allocated to a recipe food group according to the type of dish. The weight of each cooked ingredient was calculated using the raw weight recorded by the cohort member, a weight loss factor for the whole dish (from a comparable recipe in McCance and Widdowson's *The Composition of Foods series*¹) and the weight of the portion consumed. Where the food was stated as homemade but there was no recipe given, a standard homemade recipe food code was chosen.

3.5 Quality control

3.5.1 Training

HNR was responsible for providing dietary assessment training for clinic staff, who were trained prior to the start of visits with cohort members. HNR personnel demonstrated to the nurses how and where to label the diary and the accompanying zip-lock plastic wallet, where to record the appointment date, and how to use the examples and the photographs of portion sizes in the front of the diary to help the cohort members to complete the 5-day record. Emphasis was also placed on the importance of asking the cohort members to provide as much detail as possible about their intake, as well as asking them not to change their usual diet during the recording period. Nurses were

instructed to encourage the cohort members to keep the diary about food and drink consumed for 3 weekdays and 2 weekend days.

3.5.2 Dietary assessment

At the start of the coding process, editors checked ten complete diaries for each coder and gave them individual feedback on their work. For a random 10% of all diaries the editors undertook a further 100% check of all food and portion code entries. This ensured that error rates were monitored for all the coders working on the project and helped identify any coding issues. An error rate greater than 5% from any one coder resulted in further 100% checks of their coding until the editor was satisfied that problems had been eliminated. All of the entries flagged as queries by the coders were categorised into eight query types, such as food code or portion code not available in DINO, recipes, missing or insufficient detail to code food or portion.

At the end of coding and editing, each cohort member's energy and nutrient intake for each diary day was calculated. The nutrient data were examined for outliers and those identified as being either above or below the normal distribution were identified and the original diary entries checked against the coding. Portion sizes were also examined by food group for outliers and those identified checked against diary entries.

3.6 Dietary results

After making corrections the results for each day were aggregated to produce mean daily intakes. Only diaries with at least three complete days of recording were considered valid. Mean intakes were compared with Estimated Average Requirements for energy (EAR), Dietary Reference Values (DRV) for some macronutrients; Reference Nutrient Intakes (RNI) and Lower Reference Nutrient Intakes (LRNI) for vitamins and minerals.

Results were compared with the dietary results from the dietary assessment carried out in 1999 when the cohort members were 53 years old, for those who had completed a dietary record at both timepoints. However, it was not possible to compare directly the many of the food groups consumed as the structure of food groups is different at these two timepoints. The food group structure in the 1999 dataset was that used in the analysis and publication of dietary patterns in relation to chronic disease risk⁴. The 2006-11 data, on the other hand, were coded according to the now standard in-house food group structure, which provides greater flexibility for examining a variety of nutritional research questions. The same food group structure has been used in other

⁴ Mishra GD, Prynne CJ, Paul AA, Greenberg DC and Bolton-Smith C. The impact of inter-generational social and regional circumstances on dietary intake patterns of British adults: results from the 1946 British Birth Cohort. *Public Health Nutrition* 2004; 7: 737-44.

research work conducted at HNR, for example the National Diet and Nutrition Survey⁵. In this report, comments were made where it was possible to match food groups between these two assessments. For those wishing to make more detailed comparisons of dietary data from different timepoints or to examine trends in consumption of food groups, this would require the researcher to re-analyse the raw dietary data and set up food groups which suit their particular research questions and the types of analyses proposed to address these.

The differences noted for foods consumed between 1999 and 2006-11 are not necessarily statistically significant nor show trends in consumption. Dietary intakes have wide variation and differences are unlikely to be significant, and trends tend to be small but consistent over time. No statistical analyses were conducted for foods as the structures of food groups are not exactly the same between the 1999 and the 2006-11 assessments. For energy and nutrient intakes, statistical comparisons were made using Analysis of Variance (ANOVA) to test for significance, with analyses conducted using SPSS Version 14.

⁵ National Diet and Nutrition Survey: Headline results from Year 1 and Year 2 (combined) of the Rolling Programme, 2008/9 and 2009/10. Appendix A
http://www.dh.gov.uk/en/Publicationsandstatistics/Publications/PublicationsStatistics/DH_128166

4 Results

4.1 Introduction

This section reports on the collection rates for diet diaries returned from cohort members and the dietary data yielded from the valid diet diaries, defined as a diary with at least 3 days of completed dietary record. Comparisons between the 1999 and the 2006-11 dietary datasets have been made where appropriate.

4.2 Dietary data collection

1869 cohort members, 880 men and 989 women completed at least 3 days of the diet record between 2006 and 2011, representing 84% of those (83% of men and 85% of women) who attended a clinic or home visit.

Of the 1869 cohort members who completed a diary in the 2006-11 survey, 594 men (67.5%) and 724 women (73.2%) also completed at least 3 days of dietary record in 1999. Of the 1869 cohort members, 1803 (96%) completed the questions at the back of the diary which assist with diet coding. 1074 (57%) provided some food packages, the mean number being 2.87 packages per cohort member who completed a valid diary. The total number of packages received was 5368.

(Table 4.2.1)

Table 4.2.1 Numbers of diaries and additional information

Number of days recorded	n	%
<3	1	0.1
3	8	0.4
4	43	2.3
5	1818	97.2
Total diaries received	1870	100.0
Total valid diaries received (excluded diary with <3 days recorded)	1869	-
Of which:		
Questionnaire completed		
Yes	1803	96.5
No	66	3.5
Diaries returned with at least one food package		
Yes	1074	57.5
No	795	42.5
Number of packages per diary returned		
Mean	2.9	-
Range	0-34	-
Total number of packages returned	5368	-

Since the cohort members were asked to complete the diary in the days following the clinic or home visit and encouraged to complete 3 weekdays and 2 weekend days, Fridays, Saturdays and Sundays were over represented (Table 4.2.2). It is possible that the over-representation of weekend days could have led to some bias in reported food consumption and nutrient intake, since it has been shown that there is day-to-day variation in intake of some foods and nutrients, although this tends to be greater for younger adults than the age reported here.

(Table 4.2.2)

Table 4.2.2 Days of week recorded from 1869 valid diaries

Day of week	n	%
Monday	1194	12.9
Tuesday	908	9.8
Wednesday	1087	11.7
Thursday	1236	13.3
Friday	1536	16.5
Saturday	1709	18.4
Sunday	1618	17.4
Total	9288	100

4.3 Foods consumed

Results are presented for the average daily weight of food groups consumed and of the average daily intake of energy and nutrients from food sources only. Table 4.3.1 shows the average weight of food groups consumed per day (including non-consumers) and the percentage of consumers.

When comparing foods consumed between the 1999 and the 2006-11 assessments, it is not possible to compare directly the weights of food consumed as the structure of the food groups was not the same at these two timepoints. For the 1999 dataset, the food group structure was mainly dictated by the previous research interest in dietary patterns, where individual food codes were grouped in a specific way to facilitate such analyses⁶, whereas for the 2006-11 data, a standard in-house food group structure was used to provide a generic dataset. The same food group structure had been used in other research work conducted at HNR, for example the National Diet and Nutrition Survey⁷. In this report, comments were made where it was possible to match food groups between these two assessments.

⁶ Mishra GD, Prynne CJ, Paul AA, Greenberg DC and Bolton-Smith C. The impact of inter-generational social and regional circumstances on dietary intake patterns of British adults: results from the 1946 British Birth Cohort. *Public Health Nutrition* 2004; 7: 737-44.

⁷ National Diet and Nutrition Survey: Headline results from Year 1 and Year 2 (combined) of the Rolling Programme, 2008/9 and 2009/10. Appendix A
http://www.dh.gov.uk/en/Publicationsandstatistics/Publications/PublicationsStatistics/DH_128166

4.3.1 Cereals and cereal products

The subgroup of cereal and cereal products with the highest average consumption was white bread, which also had the greatest percentage of consumers in this group at 78.6% of men and 72.6% of women. Consumption of white bread was reduced from the 1999 assessment and there was a small increase in consumption of brown/granary and wheatgerm breads. Women ate less wholemeal bread than in 1999 but there was no change in men although nearly half of all cohort members consumed some wholemeal bread in 2006-11. Rice was consumed by around 40% of cohort members; there was no change in the average weight consumed from 1999. The weight of pasta consumed was around half of that reported in 1999 (from 36.7g/d to 19.3g/d in men and from 33.0g/d to 16.0g/d in women), but this was most likely due to pasta dishes being grouped differently at that assessment. For example, lasagne was coded as a pasta dish in 1999; however, this has been revised and further differentiated in 2006-11 to improve the precision of classifying composite dishes such that lasagne was by default coded as “beef, veal and dishes”, pork lasagne was coded as “pork and dishes” and vegetable lasagne coded as “other vegetables”. It is only when the recipe was available in the diet diary that the lasagne would be coded as “pasta and pasta dishes”.

Oat based and high fibre breakfast cereals were consumed in greater quantities and by more participants than low fibre breakfast cereals. There was a large reduction in the consumption of cakes and sweet puddings since 1999 but biscuit consumption was not changed. A new product since 1999, cereal bars, was consumed in greater quantities than other sweet cereal products such as cakes and biscuits and by a high proportion of cohort members (71.3% all, 68.8% men, 73.5% women).

(Table 4.3.1)

4.3.2 Milk and milk products

Semi-skimmed milk was the most commonly consumed type of milk; 80.4% of all cohort members recorded this type. Overall milk consumption was decreased from the 1999 assessment. Cheese was consumed by very high proportions (81.3% men, 83.3% women); mean consumption showed very little change since 1999 but the mean intake of yoghurt and drinking yoghurts increased with 46% of men and 62.4% of women consuming these products in 2006-11.

(Table 4.3.1)

4.3.3 Eggs and egg dishes

73% of cohort members consumed eggs in the 5 days of dietary record in 2006-11. Consumption of eggs increased slightly since 1999.

(Table 4.3.1)

4.3.4 Fats and Oils

Plant-based margarines were the most common type of spread used by cohort members, with 74.2% of men and 70.3% of women using these, compared to 50.5% of men and 56% of women using butter.

(Table 4.3.1)

4.3.5 Meat and meat products

Beef, veal and dishes⁸ and chicken, turkey and dishes were the most commonly consumed types of meat with 70.6% of men and 61.6% of women consuming beef and dishes and 72.0% of men and 70.6% of women consuming chicken and dishes, compared to 25.1% consuming lamb and dishes and 33.4% of men and 30.5% of women consuming pork and dishes. The average intakes of both red and white meats and dishes made from these had fallen since 1999. Bacon and ham was consumed by 78.1% of men and 69.9% of women but average intake was unchanged since the 1999 assessment; consumption of other processed meats and meat products including sausages, burgers, kebabs and meat pies, had risen.

(Table 4.3.1)

4.3.6 Fish and fish products

White fish was consumed by 56.9% of cohort members and oily fish by 39.8% of men and 43.6% of women over the 5 day recording period. Quantities consumed had not changed since 1999.

(Table 4.3.1)

4.3.7 Vegetables

Since 1999, average consumption of total vegetables (excluding potatoes) had risen from 156.8g to 185.0g per day for men and from 167.1g to 187.6g per day for women. 100% of cohort members ate some vegetables during the recording period in the 2006-11 assessment. Tomatoes were consumed by over 85% of cohort members and brassicas by over 70%. Over 90% ate potatoes, but the average weight consumed per day had fallen since 1999 for both men and women.

(Table 4.3.1)

4.3.8 Fruit

The average weight of fruit consumed by men had risen since 1999, while there was a slight fall in the quantity consumed by women. 91% of men and 97% of women had eaten some fruit during the reporting period in the 2006-11 assessment.

(Table 4.3.1)

⁸ "Beef, veal and dishes" refers to "beef, veal or any dishes consist of beef or veal". Similar rule applies to "chicken, turkey and dishes", "lamb and dishes", "pork and dishes", "liver and dishes", "other offal and dishes, e.g. haggis, faggots".

4.3.9 Sugar, preserves and confectionery

There was a small reduction in the average weight of sugar, preserves and confectionery consumed in 2006-11 compared with 1999. Pure sugar was the most commonly consumed component in this group, at 58.5% of men and 50.2% of women. Jam and marmalade was also consumed by many, at 48.2% of men and 50.6% of women, as was chocolate confectionery at 43.3% of men and 55.2% of women.

(Table 4.3.1)

4.3.10 Beverages

One of the food groups which can be affected by a higher proportion of weekend days is beverages, especially alcoholic beverages, which tend to be consumed more on Fridays, Saturdays and Sundays, especially by men. However, this greater consumption at weekends tends to be seen in the younger years of adulthood, rather than by those in middle age and older. Hence the distribution of days of the week is unlikely to affect the results for 2006-11 markedly but may affect comparison to earlier assessments when cohort members were younger.

Alcoholic beverages were consumed by 81.3% of men and 67.6% of women in 2006-11 over the 5 days of recording; average consumption by men had fallen since 1999 but not by women. Over one third of cohort members drank carbonated soft drinks and around two thirds, fruit based drinks. The majority of the latter was pure fruit juice or smoothies which were consumed by 45% of men and 55% of the women. Over 80% of cohort members consumed tea and coffee (tea: 87.3% of men, 87.5% of women; coffee: 84.5% of men, 83.7% of women). Coffee consumption was lower in 2006-11 compared with 1999 for both men and women, but women consumed on average more tea than in 1999 although men consumed slightly less.

(Table 4.3.1)

4.3.11 Disaggregated foods

Table 4.3.2 shows the mean daily consumption of vegetables, fruit, meat and fish after disaggregation of composite dishes. Disaggregation provides a more accurate estimate of consumption of specific food items, particularly vegetables and meat that are commonly included in composite dishes⁹. After disaggregation, consumption of vegetables was 21g/day greater for men (205.6g/d versus 185g/d) and 16g/day greater for women (203.8g/d versus 187.6g/d). Fruit consumption was greater by 10g/day for men (153.7g/d versus 143.8g/d) and 7g/day for women (168.3g/d versus 161.1g/d). Consumption of meat was 32g/day less for men and 21g/day less for women than that

⁹ Fitt E, Mak TN, Stephen AM, Prynne C, Roberts C, Swan G, Farron-Wilson M. Disaggregating composite food codes in the UK National Diet and Nutrition Survey food composition databank. *European Journal of Clinical Nutrition* 2010; 64 (Suppl 3):S32-6.

reported as meat and meat dishes above (112.1g/d versus 144.4g/d for men; 79.5g/d versus 100.3g/d for women).

When portions of fruit and vegetables were calculated according to the Department of Health guidelines⁴, it was found that the mean consumption for both men and women was just above 5 portions per day, at 5.1 and 5.3 portions respectively, and 47.8% of men and 50.9% of women consumed at least 5 portions of fruit and vegetables per day.

(Table 4.3.2)

4.4 Dietary supplements

Table 4.4 shows the percentage of men and women taking particular types of dietary supplements in 2006-11 during the 5 day period of dietary recording. The most commonly taken supplements, taken by 24.2% of men and 32.7% of women were fish oils including cod liver oil. Multivitamins and minerals were taken by 7% of men and 15.2% of women. Other supplements that did not fall into any of the defined categories (e.g. cranberry extract, flaxseed) were taken by 13.4% of men and 24.1 % of women. In 1999, 38.3% of cohort members recorded at least one supplement during the 5 days. In 2006-11, this percentage was 43.9%; 33.8% men and 53.0% women. It should be noted that supplements taken on a less frequent basis may not be captured in the 5 days of dietary intake recording, and hence these percentages may not reflect overall supplement consumption by cohort members.

(Table 4.4)

4.5 Energy and nutrient intakes

4.5.1 Energy and macronutrient intakes

Table 4.5.1 shows the mean daily intakes of energy and macronutrients from food sources for all the 2006-11 cohort members who completed at least 3 days of dietary record. Table 4.5.2 shows the percentages of cohort members who met the Dietary Reference Values for energy and some macronutrients in 2006-11. Tables 4.5.7a, 4.5.7b and 4.5.7c show mean daily intakes of energy and macronutrients from food sources for the 1318 cohort members who recorded their diet both in 1999 and 2006-11. Changes in energy and nutrient intakes refer to the latter subsample.

In 2006-11 energy intakes of men and women were, on average 87.5% and 88.8% of the Average Energy Requirement (AER) for men and women of this age (Table 4.5.2); the intakes ranged from 4.9-12.4 MJ/day for men and 4.2-10.1 MJ/day for women. Both men and women had significantly lower energy intakes in 2006-11 compared with 1999. Total fat intake (g/d) was significantly lower for men in 2006-11 (86.5g/d versus 80.6g/d, $p < 0.001$, Table 4.5.7a) but not for women (68.1g/d versus 66.3g/d, $p = \text{NS}$, Table 4.5.7b); average saturated fat intake (g/d) was significantly lower in men (34.2g/d

versus 30.1g/d, $p < 0.001$, Table 4.5.7a) and women (26.5g/d versus 25g/d, $p = 0.004$, Table 4.5.6b). The percentage of energy from fat did not decrease and 54.1% of men and 57.9% of women had energy from fat above the recommendation of 33% total energy (Table 4.5.2). Energy from saturated fat did fall; in 2006-11 79.0% of men and 79.9% of women were above the recommended 10% of total energy (Table 4.5.2), compared to 86.2% of men and 85.2% of women in 1999. Intakes of monounsaturated fatty acids did not change but intakes of trans fatty acids were significantly reduced in 2006-11 compared to 1999 (Table 4.5.7c). There was small significant rise in the intake of n-3 polyunsaturated fatty acids and a small significant fall in the intake of n-6 polyunsaturated fatty acids (Table 4.5.7c). Protein intakes did not change but intakes of carbohydrate, particularly starch, were significantly lower (Table 4.5.7c). In 2006-11, 70.6% of men and 60.8% of women did not reach the recommended 47% of total energy from carbohydrate (Table 4.5.2). Intakes of non-starch polysaccharide were significantly higher in 2006-11 compared with 1999 (16.1g/d versus 14.9g/d, $p < 0.001$, Table 4.5.7c). Intakes of alcohol were not significantly different between timepoints (Table 4.5.7c). Men consumed just over 7% energy from alcohol at both timepoints (Table 4.5.7a) and women 4% (Table 4.5.7b).

4.5.2 Intakes of vitamins and minerals

Tables 4.5.3 and 4.5.4 show mean daily intakes of vitamins and minerals from food sources only and from foods plus dietary supplements. Tables 4.5.5 and 4.5.6 show mean intakes expressed as a percentage of the Reference Nutrient Intake (RNI) and the percentage of cohort members failing to achieve the Lower Reference Nutrient Intake (LRNI) for micronutrients from food sources only and from food plus supplements. Tables 4.5.8a, 4.5.8b and 4.5.8c show the same results for micronutrients from food sources only for cohort members included in both the 1999 and 2006-11 surveys. Table 4.5.9a, 4.5.9b and 4.5.9c show mean intakes of the 1999 and the 2006-11 cohorts expressed as a percentage of the RNI and the percentage of cohort members failing to achieve the LRNI for micronutrients from food sources plus supplements.

Intakes of several vitamins and minerals were greater in 2006-11 compared to intakes in 1999. Niacin, vitamin B₆ and folate were significantly higher in 2006-11 (Table 4.5.8c); intakes of thiamine and vitamin C were significantly higher in men (Table 4.5.8a) whereas intakes of vitamin B₁₂ and D were significantly higher in women (Table 4.5.8b). Vitamin E intakes were lower for both men and women (Table 4.5.8c). In 2006-11, mean intakes of the vitamins reported for both men and women were all above the RNI and the intakes of very few cohort members fell below the LRNI for vitamin A, riboflavin and folate (Table 4.5.5). A comparison of the 2006-11 with the 1999 data shows that

the percentage of cohort members with intakes below the LRNI has fallen since 1999 for majority of the micronutrients, except for riboflavin (Table 4.5.9c).

When micronutrients from supplements were included, average intakes in 2006-11 of some vitamins were raised by more than 35%; intakes of vitamin E, thiamine, riboflavin and vitamins D and C were raised by 58%, 54%, 42%, 50% and 42% respectively (Table 4.5.3 and Table 4.5.4).

Intakes of several minerals: iron, non haem iron, magnesium, copper and zinc were significantly greater in 2006-11 compared with 1999 while intakes of calcium and sodium were reduced (Table 4.5.8c). In 2006-11, the mean intakes of all reported minerals were above the RNI for men and women with the exception of potassium and copper intakes for women (Table 4.5.5). A small percentage of cohort members had mineral intakes below the LRNI, the greatest being 4.5% of men below the LRNI for magnesium and 4.6% of women below the LRNI for potassium (Table 4.5.5). Comparing 2006-11 with 1999, the percentage of men and women with intakes of potassium and zinc below the LRNI had risen in 2006-11 (Table 4.5.9c); more women were below the LRNI for calcium (Table 4.5.9b) and more men, for magnesium (Table 4.5.9a).

Supplements taken in 2006-11 had less impact on average intakes of minerals compared to vitamins. Zinc intakes were raised by 14%, copper intakes, 11%, iron intakes, 8% and calcium, 6% (Table 4.5.3 and Table 4.5.4).

Although average intakes of micronutrients from all sources as a percentage of RNI were raised in comparison with intakes from food only, there was little impact on the percentage of cohort members falling below the LRNI (Table 4.5.5 and Table 4.5.6).

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Table 4.3.1 Average daily consumption of food groups (grams) of men and women aged 60-64 years (2006-11): all cohort members including non consumers

	Men (n=880)			Women (n=989)			All (n=1869)		
	Mean	SD	% consumers	Mean	SD	% consumers	Mean	SD	% consumers
Cereals and Cereal products									
Pasta and pasta dishes	19.3	31.5	37.2	16.0	26.4	36.3	17.5	28.9	36.7
Rice and rice dishes	20.2	28.7	43.0	15.9	25.0	38.6	17.9	26.9	40.7
Pizza	6.1	19.7	11.9	4.1	14.2	10.5	5.1	17.0	11.2
Other cereals and dishes	8.8	15.1	51.3	8.1	11.5	57.1	8.5	13.3	54.4
White bread	46.3	44.5	78.6	28.3	30.8	72.6	36.8	38.9	75.4
Brown/granary/wheatgerm bread	15.6	31.0	36.0	11.9	21.9	38.4	13.6	26.6	37.3
Wholemeal bread	27.4	40.5	48.1	20.5	28.7	52.3	23.7	34.9	50.3
Other bread	8.9	19.0	32.0	8.0	16.5	33.9	8.4	17.7	33.0
Crispbreads, e.g. rivita	0.9	3.8	8.3	1.3	4.0	14.7	1.1	3.9	11.7
Oat based breakfast cereals	18.2	36.4	40.7	21.1	37.3	51.2	19.7	36.9	46.2
Other breakfast cereals, high fibre	8.4	15.0	35.2	6.5	12.3	33.2	7.4	13.6	34.1
Other breakfast cereals, low fibre	6.2	11.7	34.1	6.4	19.2	33.4	6.3	16.1	33.7
Biscuits	9.5	14.2	57.2	8.1	10.6	64.6	8.8	12.5	61.1
Cereal bars	28.4	32.0	68.8	25.1	26.8	73.5	26.6	29.4	71.3
Cereal based puddings (not milk)	1.0	4.4	7.7	1.3	4.3	11.9	1.1	4.4	10.0
Milk based puddings	4.2	14.4	11.4	3.5	11.5	10.9	3.8	13.0	11.1
Cakes, pastries, buns and pies	7.5	21.1	18.3	5.9	17.2	16.3	6.7	19.1	17.2

	Men (n=880)			Women (n=989)			All (n=1869)		
	Mean	SD	% consumers	Mean	SD	% consumers	Mean	SD	% consumers
Milk and milk products									
Whole milk	21.8	83.4	16.6	13.0	56.3	13.4	17.2	70.5	14.9
Semi-skimmed milk	141.6	129.5	81.3	123.1	127.3	79.6	131.8	128.7	80.4
One percent milk	3.6	32.3	1.9	1.6	19.6	1.3	2.6	26.4	1.6
Skimmed milk	32.3	89.0	27.3	48.1	93.7	35.7	40.6	91.8	31.7
Other milk, animal based	1.3	19.3	0.7	1.7	19.9	1.3	1.5	19.6	1.0
Other milk, plant based	4.7	32.7	3.6	10.3	46.0	8.1	7.7	40.4	6.0
Milk based drinks	0.2	3.9	0.5	0.3	6.7	0.2	0.2	5.6	0.3
Cream and fromage frais	4.8	11.1	30.1	4.7	9.2	38.3	4.7	10.1	34.5
Yoghurt and drinking yoghurts	34.6	51.2	46.0	47.6	55.5	62.4	41.5	53.9	54.7
Ice cream and dairy desserts	14.3	22.6	42.4	12.8	20.1	45.5	13.5	21.3	44.0
Cheese including cottage cheese	17.0	16.8	81.3	15.8	17.8	83.3	16.3	17.4	82.3
Egg and egg dishes	22.8	24.3	73.2	19.8	20.9	72.9	21.2	22.6	73.0
Fats and oils									
Butter	6.2	10.2	50.5	5.4	8.3	56.0	5.8	9.3	53.4
Animal based fats (solid)	0.2	0.9	5.7	0.2	0.8	6.9	0.2	0.8	6.3
Plant based fats (solid)	11.2	11.1	74.2	7.4	8.3	70.3	9.2	9.9	72.1
Oils	2.4	3.1	66.3	2.1	2.7	69.9	2.2	2.9	68.2

	Men (n=880)			Women (n=989)			All (n=1869)		
	Mean	SD	% consumers	Mean	SD	% consumers	Mean	SD	% consumers
Meat and meat dishes and products									
Beef, veal and dishes	40.2	43.7	70.6	28.0	34.0	61.6	33.7	39.4	65.8
Lamb and dishes	9.1	19.6	25.1	6.9	14.9	25.2	7.9	17.3	25.1
Pork and dishes	10.4	18.5	33.4	7.6	13.4	30.5	9.0	16.0	31.9
Other red meat, e.g. rabbit, venison	0.2	1.9	0.9	0.4	3.5	1.2	0.3	2.8	1.1
Chicken, turkey and dishes	38.0	41.4	72.0	30.7	32.4	70.6	34.1	37.1	71.3
Other white meat e.g. duck, pheasant	1.5	7.9	5.0	1.2	6.5	4.3	1.3	7.2	4.7
Bacon and ham	19.1	21.0	78.1	12.7	15.6	69.9	15.7	18.6	73.7
Other processed meats	0.9	4.6	7.6	0.5	2.4	5.6	0.7	3.6	6.5
Processed meat pies	9.9	20.3	27.2	4.2	10.8	16.7	6.9	16.2	21.6
Sausages, burgers and kebab	12.1	17.4	46.1	6.0	11.3	31.3	8.9	14.8	38.3
Liver and dishes	2.4	9.2	9.9	1.4	5.8	8.6	1.8	7.6	9.2
Other offal and dishes, e.g. haggis, faggots	0.6	4.2	3.4	0.7	4.5	3.5	0.6	4.4	3.5
Fish and fish dishes									
White fish, including canned tuna	20.8	23.7	56.8	18.1	21.4	56.9	19.4	22.5	56.9
Oily fish	12.6	20.4	39.8	12.2	18.4	43.6	12.4	19.4	41.8
Shellfish	4.7	12.9	23.1	5.2	12.0	26.6	5.0	12.5	24.9
Vegetables and potatoes									
Brassicacea	26.8	29.0	70.2	28.4	30.3	77.0	27.7	29.7	73.8
Tomatoes	37.0	35.3	85.2	39.4	34.5	87.4	38.3	34.9	86.4
Yellow, red and dark green leafy vegetables	7.6	13.1	45.5	9.9	14.9	56.4	8.8	14.1	51.3
Other vegetables	97.5	60.8	98.9	96.9	55.9	99.5	97.2	58.3	99.2
Pulses and lentils	3.8	12.0	16.1	4.3	13.5	18.7	4.1	12.8	17.5
Baked beans	12.2	23.2	30.1	8.6	17.7	26.3	10.3	20.6	28.1
Potatoes	77.9	52.5	90.8	70.4	46.3	91.7	73.9	49.4	91.3
Potato products, e.g. chips, wedges, instant mash	35.4	37.3	65.8	25.7	30.2	57.3	30.3	34.1	61.3
Fruit									
Fresh fruit	130.2	112.0	89.5	145.3	103.6	95.1	138.2	107.9	92.5
Canned and cooked fruit	10.2	25.0	28.2	11.2	23.0	34.4	10.7	24.0	31.5
Dried fruit	3.4	12.9	21.6	4.6	10.8	34.1	4.0	11.9	28.2

	Men (n=880)			Women (n=989)			All (n=1869)		
	Mean	SD	% consumers	Mean	SD	% consumers	Mean	SD	% consumers
Nuts and Seeds	6.0	14.2	38.6	6.1	11.8	44.5	6.0	13.0	41.7
Savoury snacks									
Cereal based snacks	0.5	2.7	6.9	0.3	1.9	5.5	0.4	2.3	6.2
Potato based snacks	3.4	7.1	31.8	2.5	6.1	30.9	2.9	6.6	31.4
Vegetable based snacks	0.0	0.8	0.7	0.0	0.5	0.6	0.0	0.6	0.6
Savoury biscuits and crackers	2.7	7.0	27.5	2.9	5.9	34.0	2.8	6.4	30.9
Sugar, preserves and confectionery									
Pure sugar	9.1	15.1	58.5	4.5	9.2	50.2	6.7	12.6	54.1
Other sugar, e.g. syrups, honey	2.0	5.4	23.4	1.7	4.1	26.3	1.8	4.7	24.9
Jam and marmalade	5.7	9.0	48.2	4.4	7.2	50.6	5.0	8.1	49.4
Chutney and pickles	3.2	6.7	31.0	3.3	8.6	30.2	3.3	7.7	30.6
Sugar confectionery	2.1	8.4	13.9	3.2	11.6	24.2	2.7	10.2	19.3
Chocolate confectionery	5.9	10.1	43.3	7.4	11.2	55.2	6.7	10.7	49.6
Sorbets and lollies	0.3	2.7	1.8	0.4	3.4	2.1	0.4	3.1	2.0
Beverages									
Alcohol	406.7	515.0	81.3	116.6	163.0	67.6	253.2	399.8	74.1
Carbonated soft drinks	243.0	508.3	35.6	213.0	526.5	38.9	227.1	518.1	37.3
Fruit based drinks	91.2	137.6	62.6	72.6	91.4	68.0	81.4	115.8	65.5
Tea	562.2	433.3	87.3	594.0	410.4	87.5	579.0	421.5	87.4
Coffee	346.9	343.0	84.5	313.1	315.8	83.7	329.0	329.2	84.1
Water, still, tap, sparkling, flavoured	230.8	346.6	72.3	406.1	410.5	88.8	323.6	391.6	81.0
Powdered beverages, e.g cocoa, horlicks	7.9	34.8	16.3	13.2	43.7	24.6	10.7	39.8	20.7
Miscellaneous									
Canned, fresh and homemade soups	26.8	48.6	37.3	23.7	41.5	37.2	25.2	45.0	37.2
Dried soups	3.8	19.5	4.5	4.8	28.3	6.0	4.3	24.5	5.3
Cooking sauces, e.g. gravies, pesto, cooking sauces	16.7	21.9	61.0	13.1	17.3	60.1	14.8	19.7	60.5
Dressings and mayonnaise	2.7	5.0	41.7	3.5	5.3	53.6	3.1	5.2	48.0
Other savoury sauces, e.g. brown sauce, ketchup	5.0	8.6	50.6	5.0	8.9	53.7	5.0	8.7	52.2
Dried herbs and spices	0.7	2.0	44.8	0.6	1.4	48.8	0.7	1.7	46.9

Table 4.3.2 Average daily consumption of disaggregated foods of men and women age 60-64 (2006-2011): all cohort members including non consumers

	Men (n=880)		Women (n=989)		All (n=1869)	
	Mean	SD	Mean	SD	Mean	SD
Total vegetables (incl from composite dishes) g/d	205.6	99.9	203.8	91.9	204.6	95.7
Total fruit (incl from composite dishes) g/d	153.7	118.4	168.3	107.8	161.5	113.1
Total fruit and vegetables (incl from composite dishes) g/d	359.3	176.6	372.1	163.2	366.1	169.7
Total portions of fruit and vegetables g/d	5.1	2.5	5.3	2.3	5.2	2.4
% achieving 5 portions/day	47.8	-	50.9	-	49.4	-
Total red meat (incl from composite dishes) g/d	81.0	45.6	53.3	34.2	66.4	42.3
Total white meat (incl from composite dishes) g/d	31.0	30.9	26.2	25.4	28.5	28.2
Total meat (incl from composite dishes) g/d	112.1	50.5	79.5	39.1	94.8	47.7
Total fish (incl from composite dishes) g/d	30.1	26.4	28.4	24.9	29.2	25.6

Table 4.4 - Percent of cohort members, men and women aged 60-64 years (2006-2011) taking dietary supplements

	Men %	Women %	All %
Cod liver oil and other fish oils	24.2	32.7	28.7
Evening primrose oil and other plant oils	1.5	8.2	5.0
Single vitamins/minerals NOT Folic acid, iron, calcium, vitamin D	4.8	8.6	6.8
Folic acid	0.7	1.1	0.9
Iron only or with vitamin C	0.1	0.8	0.5
Calcium only or with vitamin D	1.3	5.9	3.7
Vitamins (two or more including multivitamins) no minerals	1.8	3.8	2.9
Minerals (two or more including multi-minerals) no vitamins	0.9	0.8	0.9
Vitamins and minerals Including multivitamins	7.0	15.2	11.3
Non-nutrient supplements (including herbal)	8.3	13.5	11.1
Other supplements	13.4	24.1	19.0
<i>Base</i>	<i>880</i>	<i>989</i>	<i>1869</i>

Table 4.5.1 Average daily intake of energy and macronutrients from food sources of men and women aged 60-64 years (2006-2011)

	Men (n=880)					Women (n=989)					All (n=1869)				
	Mean	Median	SD	Upper 2.5 %ile	Lower 2.5 %ile	Mean	Median	SD	Upper 2.5 %ile	Lower 2.5 %ile	Mean	Median	SD	Upper 2.5 %ile	Lower 2.5 %ile
Total energy															
MJ	8.8	8.8	1.9	12.4	4.9	7.1	7.0	1.5	10.1	4.2	7.9	7.8	1.9	11.9	4.4
kcal	2083	2082	465	2958	1152	1687	1670	361	2404	1002	1873	1848	458	2842	1041
Food energy															
MJ	8.1	8.1	1.8	11.9	4.5	6.8	6.7	1.5	9.8	4.1	7.4	7.3	1.8	11.0	4.2
kcal	1931	1921	440	2827	1068	1618	1599	347	2327	970	1765	1733	423	2625	997
Protein															
g	83.9	82.8	18.6	122.1	48.7	70.5	70.0	14.8	100.9	42.6	76.8	75.4	18.0	114.1	44.5
% food energy	17.7	17.4	3.1	24.5	12.6	17.7	17.4	3.4	25.3	12.2	17.7	17.4	3.2	25.0	12.4
% total energy	16.4	16.0	2.8	23.0	11.8	17.0	16.7	3.3	24.3	11.6	16.7	16.4	3.1	23.8	11.7
Fat															
g	78.3	75.8	23.9	132.1	38.1	64.4	62.7	19.7	107.4	28.1	70.9	68.8	22.9	119.4	31.8
% food energy	36.2	36.1	5.6	47.1	25.2	35.5	35.6	6.2	47.9	23.1	35.8	35.8	5.9	47.4	23.6
% total energy	33.6	33.6	5.7	43.9	22.8	34.0	34.2	6.1	46.3	21.9	33.8	33.9	5.9	45.3	22.4
Saturated fatty acids															
g	29.3	28.1	10.6	52.2	12.3	24.3	22.9	9.0	44.7	9.5	26.7	25.2	10.1	49.1	10.5
% food energy	13.5	13.4	3.2	20.6	7.9	13.4	13.2	3.4	20.5	7.0	13.4	13.3	3.3	20.6	7.4
% total energy	12.6	12.4	3.1	19.5	7.3	12.8	12.8	3.3	20.3	6.7	12.7	12.6	3.2	19.8	7.0
Cis mono-unsaturated fatty acids															
g	26.6	25.8	8.6	45.3	12.2	21.4	20.8	7.0	36.7	9.0	23.8	23.0	8.2	42.4	9.8
% food energy	12.3	12.2	2.4	17.5	8.2	11.8	11.8	2.5	16.9	7.1	12.0	12.0	2.4	17.1	7.4
% total energy	11.4	11.3	2.3	16.3	7.4	11.3	11.4	2.4	16.1	6.7	11.4	11.3	2.3	16.2	7.1
Cis n-3 poly-unsaturated fatty acids															
g	2.1	1.9	0.9	4.4	0.8	1.8	1.6	0.7	3.6	0.8	1.9	1.8	0.8	4.0	0.8
% food energy	1.0	0.9	0.4	1.9	0.5	1.0	0.9	0.3	1.8	0.5	1.0	0.9	0.3	1.8	0.5
% total energy	0.9	0.8	0.3	1.7	0.4	0.9	0.9	0.3	1.8	0.5	0.9	0.9	0.3	1.8	0.5

	Men (n=880)					Women (n=989)					All (n=1869)				
	Mean	Median	SD	Upper 2.5 %ile	Lower 2.5 %ile	Mean	Median	SD	Upper 2.5 %ile	Lower 2.5 %ile	Mean	Median	SD	Upper 2.5 %ile	Lower 2.5 %ile
Cis n-6 poly-unsaturated fatty acids															
g	11.4	10.7	4.5	21.4	5.0	9.6	8.9	3.9	19.1	4.0	10.5	9.7	4.3	20.5	4.3
% food energy	5.3	5.1	1.6	8.8	2.9	5.3	5.0	1.7	9.3	2.8	5.3	5.0	1.6	9.1	2.9
% total energy	4.9	4.7	1.5	8.3	2.6	5.1	4.8	1.7	9.2	2.7	5.0	4.7	1.6	8.6	2.7
Trans fatty acids															
g	2.5	2.3	1.2	5.3	0.8	2.0	1.9	0.9	4.1	0.5	2.2	2.1	1.1	4.7	0.6
% food energy	1.1	1.1	0.4	2.1	0.5	1.1	1.1	0.4	2.1	0.3	1.1	1.1	0.4	2.1	0.4
% total energy	1.0	1.0	0.4	2.1	0.4	1.0	1.0	0.4	2.0	0.3	1.0	1.0	0.4	2.0	0.4
Carbohydrate															
g	237.5	235.5	62.8	372.4	113.0	201.8	199.9	50.6	309.8	106.2	218.6	213.7	59.4	342.1	108.4
% food energy	46.1	46.2	6.0	57.0	34.1	46.8	46.9	6.0	58.0	34.6	46.5	46.6	6.0	57.6	34.3
% total energy	42.9	43.1	7.1	55.8	27.9	45.0	45.2	6.6	56.9	31.0	44.0	44.3	6.9	56.5	30.0
Starch															
g	131.0	129.6	34.7	202.0	64.7	106.7	105.6	30.2	162.4	56.4	118.1	115.5	34.6	187.1	58.6
% food energy	25.7	25.5	4.8	34.8	16.3	24.9	24.8	5.0	34.2	15.2	25.2	25.2	4.9	34.6	15.8
% total energy	23.9	23.6	5.1	34.6	13.5	23.9	23.9	5.1	33.7	14.0	23.9	23.8	5.1	34.1	13.8
Sugars															
g	104.5	101.5	40.3	188.6	31.6	93.7	90.9	32.3	167.9	36.3	98.8	95.1	36.6	180.9	34.1
% food energy	20.1	20.1	5.6	30.3	9.0	21.6	21.4	5.5	32.9	11.3	20.9	20.8	5.6	32.0	9.7
% total energy	18.7	18.7	5.5	29.4	8.1	20.8	20.6	5.5	32.2	10.4	19.8	19.7	5.6	31.2	9.1
Alcohol															
g	22.3	16.5	23.1	83.1	0.0	10.2	5.4	14.0	43.2	0.0	15.9	9.3	19.8	67.0	0.0
% total energy	7.3	5.5	7.2	25.2	0.0	4.1	2.3	5.2	16.7	0.0	5.6	3.7	6.4	22.9	0.0
Non starch polysaccharide															
g	16.3	15.8	5.4	29.0	7.1	15.0	14.6	4.7	25.3	6.9	15.6	15.1	5.1	27.1	7.0

Table 4.5.2 Macronutrient intake of men and women aged 60-64 years (2006-2011) compared to Dietary Reference Values (DRVs)

Macronutrients	DRV			Men	Women
Energy; AER MJ/day	9.93 (men)	7.99 (women)	Percent of AER	87.5	88.8
Total fat % total energy	33		Percent above DRV	54.1	57.9
Saturated fat % total energy	10		Percent above DRV	79.0	79.9
Total carbohydrate % energy	47		Percent below DRV	70.6	60.8
			<i>Base</i>	<i>880</i>	<i>989</i>

AER=Average Energy Requirement

Table 4.5.3 Average daily intake of vitamins and minerals from food sources of men and women aged 60-64 years (2006-2011)

	Men (n=880)					Women (n=989)					All (n=1869)				
	Mean	Median	SD	Upper 2.5 %ile	Lower 2.5 %ile	Mean	Median	SD	Upper 2.5 %ile	Lower 2.5 %ile	Mean	Median	SD	Upper 2.5 %ile	Lower 2.5 %ile
Vitamins															
Retinol µg	683	374	1285	5120	121	509	318	843	2886	98	591	340	1077	4100	113
Carotene equivalent. µg	2708	2358	1797	6779	441	2778	2540	1647	6928	495	2745	2438	1719	6841	459
Thiamin mg	1.8	1.6	1.6	2.8	0.8	1.5	1.4	0.8	2.6	0.8	1.6	1.5	1.3	2.7	0.8
Riboflavin mg	2.0	1.9	0.7	3.5	0.9	1.8	1.7	0.6	3.0	0.8	1.9	1.8	0.7	3.3	0.8
Niacin mg	23.7	23.0	7.7	40.9	11.9	18.9	18.5	5.9	31.0	9.2	21.2	20.3	7.2	37.5	10.0
Pyridoxine mg	2.7	2.5	0.9	4.5	1.3	2.2	2.1	0.7	3.6	1.1	2.4	2.3	0.9	4.2	1.2
Vitamin B12 µg	7.0	5.9	4.5	21.2	2.4	5.9	5.2	3.6	15.1	1.8	6.4	5.6	4.1	18.3	2.1
Folate µg	339	331	104	560	161	290	282	97	479	143	313	303	103	538	151
Vitamin C mg	101.5	90.7	56.3	242.9	26.1	102.9	94.7	48.7	212.1	25.8	102.2	93.4	52.4	226.6	26.0
Vitamin D µg	4.0	3.4	2.6	10.1	1.1	3.4	3.0	2.1	8.4	0.7	3.7	3.2	2.3	9.7	0.9
Vitamin E alpha tocopherol equivalent mg	10.1	9.5	3.9	19.1	4.1	9.1	8.5	3.5	16.6	3.6	9.5	9.0	3.8	18.2	3.7
Minerals															
Iron mg	13.1	12.8	3.8	21.5	6.5	11.4	11.3	3.4	18.6	5.7	12.2	11.8	3.6	20.0	6.0
Haem iron mg	0.8	0.7	0.5	2.2	0.2	0.6	0.5	0.4	1.7	0.0	0.7	0.6	0.5	1.9	0.1
Non haem iron mg	12.3	12.0	3.6	20.3	5.7	10.8	10.6	3.3	17.9	5.5	11.5	11.2	3.5	18.8	5.5
Calcium mg	947	928	288	1624	464	858	838	245	1398	439	900	876	270	1494	451
Magnesium mg	326	320	90	514	168	284	280	74	438	156	303	294	84	491	162
Sodium mg	2766	2699	768	4499	1440	2200	2128	679	3675	1123	2467	2382	775	4158	1195
Potassium mg	3542	3518	876	5433	1931	3199	3170	742	4693	1788	3360	3302	826	5049	1863
Phosphorus mg	1446	1431	343	2161	798	1254	1247	279	1841	747	1344	1318	325	2014	770
Copper mg	1.41	1.23	0.88	4.28	0.64	1.19	1.08	0.60	2.51	0.53	1.29	1.15	0.75	3.26	0.57
Zinc mg	10.3	10.1	2.7	15.9	5.9	8.5	8.5	2.1	13.3	4.8	9.4	9.0	2.5	14.7	5.1

Table 4.5.4 Average daily intake of vitamins and minerals from all sources (foods and supplements) of men and women aged 60-64 years (200

	Men (n=880)					Women (n=989)					All (n=1869)				
	Mean	Median	SD	Upper 2.5 %ile	Lower 2.5 %ile	Mean	Median	SD	Upper 2.5 %ile	Lower 2.5 %ile	Mean	Median	SD	Upper 2.5 %ile	Lower 2.5 %ile
Vitamins															
Retinol µg	833	440	1333	5120	124	719	400	940	3404	120	773	418	1143	4184	124
Carotene equivalents µg	2732	2362	1826	6949	441	2835	2562	1756	7134	547	2786	2449	1789	7080	459
Thiamin mg	2.3	1.7	5.7	4.0	0.8	2.7	1.5	8.0	11.2	0.8	2.5	1.6	7.0	6.8	0.8
Riboflavin mg	2.5	2.0	5.6	4.5	0.9	2.8	1.8	6.9	7.2	0.8	2.7	1.9	6.3	5.6	0.9
Niacin mg	25.3	23.5	10.7	45.9	11.9	22.3	19.6	12.1	54.9	9.4	23.7	21.3	11.6	50.8	10.4
Pyridoxine mg	3.2	2.6	5.9	5.7	1.3	3.5	2.3	8.0	12.0	1.1	3.4	2.4	7.0	7.1	1.2
Vitamin B12 µg	9.9	6.0	37.3	25.4	2.6	8.2	5.5	22.7	26.7	2.0	9.0	5.7	30.4	26.5	2.2
Folate µg	358	339	130	672	162	341	301	265	706	144	349	319	213	700	155
Vitamin C mg	125.3	96.6	139.4	357.7	26.1	162.3	106.2	240.2	906.0	27.4	144.9	102.3	200.0	672.6	26.4
Vitamin D µg	5.3	4.1	4.0	15.3	1.2	5.9	4.1	5.3	22.6	0.9	5.6	4.1	4.8	19.2	1.0
Vitamin E alpha tocopherol equivalent mg	13.4	10.4	25.3	35.5	4.2	16.6	9.7	36.5	51.8	3.7	15.1	10.1	31.8	45.9	3.9
Minerals															
Iron mg	13.6	13.1	4.7	24.7	6.5	12.8	11.6	9.7	27.6	5.8	13.2	12.1	7.8	26.4	6.0
Haem iron mg	0.8	0.7	0.5	2.2	0.2	0.6	0.5	0.4	1.7	0.0	0.7	0.6	0.5	1.9	0.1
Non haem iron mg	12.3	12.0	3.6	20.3	5.7	10.8	10.6	3.3	17.9	5.5	11.5	11.2	3.5	18.8	5.5
Calcium mg	970	944	308	1726	479	945	870	398	1978	444	957	909	359	1852	456
Magnesium mg	331	324	96	529	168	300	283	116	518	156	314	301	108	529	163
Sodium mg	2766	2699	768	4499	1440	2200	2128	679	3675	1123	2467	2382	775	4158	1195
Potassium mg	3549	3535	875	5444	1931	3205	3174	744	4693	1788	3367	3308	826	5049	1863
Phosphorus mg	1452	1435	343	2170	801	1262	1259	285	1868	747	1352	1328	328	2037	770
Copper mg	1.53	1.24	1.96	4.54	0.64	1.34	1.10	1.15	3.47	0.54	1.43	1.17	1.59	4.02	0.57
Zinc mg	11.1	10.4	4.6	25.9	6.0	10.2	8.7	5.7	25.4	4.9	10.6	9.4	5.2	25.5	5.2

Table 4.5.5 Average intakes of vitamins and minerals from food sources of men and women aged 60-64 years (2006-2011) as percentage of Reference Nutrient Intake (RNI) and percent of cohort members below Lower Reference Nutrient Intake (LRNI)

	Men (n=880)				Women (n=989)				All (n=1869)			
	Mean % RNI	Median	SD	%below LRNI	Mean % RNI	Median	SD	%below LRNI	Mean % RNI	Median	SD	%below LRNI
Vitamins												
Vitamin A	161	114	188	2.4	161	129	146	1.0	161	123	167	1.7
Thiamin	197	182	183	0.0	190	179	96	0.1	193	180	144	0.1
Riboflavin	153	146	52	1.5	160	154	55	2.6	157	150	54	2.1
Niacin	148	144	48	0.0	157	154	49	0.3	153	149	49	0.2
Pyridoxine	190	180	68	0.0	181	177	57	0.0	185	179	62	0.0
Vitamin B12	468	395	301	0.0	395	347	240	0.2	429	370	273	0.1
Folate	169	166	52	0.3	145	141	49	0.6	157	152	52	0.5
Vitamin C	254	227	141	0.1	257	237	122	0.2	256	234	131	0.2
Minerals												
Iron	150	147	43	0.6	131	129	39	1.1	140	136	42	0.9
Calcium	135	133	41	1.1	123	120	35	1.6	129	125	39	1.4
Magnesium	109	107	30	4.5	105	104	27	2.3	107	105	29	3.4
Potassium	101	101	25	2.7	91	91	21	4.6	96	94	24	3.7
Phosphorus	263	260	62	No LRNI	228	227	51	No LRNI	244	240	59	No LRNI
Copper	118	102	74	No LRNI	99	90	50	No LRNI	108	96	63	No LRNI
Zinc	108	107	28	1.8	122	121	29	1.1	115	114	30	1.4

Table 4.5.6 Average intakes of vitamins and minerals from all sources (foods and supplements) of men and women aged 60-64 years (2006-2011) as percentage of Reference Nutrient Intake (RNI) and percent of cohort members below Lower Reference Nutrient Intake

	Men (n=880)				Women (n=989)				All (n=1869)			
	Mean % RNI	Median	SD	% below LRNI	Mean % RNI	Median	SD	% below LRNI	Mean % RNI	Median	SD	% below LRNI
Vitamins												
Vitamin A	183	129	195	2.0	198	153	164	0.7	191	140	180	1.3
Thiamin	254	187	637	0.0	341	190	999	0.1	300	188	849	0.1
Riboflavin	191	152	434	1.4	255	164	629	2.4	225	157	547	1.9
Niacin	158	147	67	0.0	186	164	101	0.2	173	156	88	0.1
Pyridoxine	231	186	418	0.0	294	188	663	0.0	264	187	562	0.0
Vitamin B12	658	401	2484	0.0	544	363	1512	0.1	598	383	2029	0.1
Folate	179	170	65	0.3	170	150	133	0.6	174	159	106	0.5
Vitamin C	313	242	348	0.1	406	265	600	0.2	362	256	500	0.2
Minerals												
Iron	156	151	55	0.6	148	133	111	1.1	152	139	89	0.9
Calcium	139	135	44	0.9	135	124	57	1.2	137	130	51	1.1
Magnesium	110	108	32	4.5	111	105	43	2.2	111	106	38	3.3
Potassium	101	101	25	2.7	92	91	21	4.4	96	95	24	3.6
Phosphorus	264	261	62	no LRNI	230	229	52	no LRNI	246	241	60	no LRNI
Copper	128	103	163	no LRNI	112	92	96	no LRNI	119	98	132	no LRNI
Zinc	117	109	48	1.8	146	125	82	1.1	132	117	70	1.4

Table 4.5.7a Average daily intake of energy and macronutrients from food sources of men aged 53 years (1999) and 60-64 years (2006-2011)

	Men 1999 (n=594)					Men 2006-2011 (n=594)					P
	Mean	Median	SD	Upper 2.5 %ile	Lower 2.5 %ile	Mean	Median	SD	Upper 2.5 %ile	Lower 2.5 %ile	
Total energy											
MJ	9.5	9.5	2.0	13.6	5.3	9.0	9.0	2.0	12.6	5.2	<0.001
kcal	2264	2267	478	3263	1265	2139	2137	468	3003	1222	<0.001
Food energy											
MJ	8.8	8.8	1.9	12.7	4.9	8.3	8.3	1.9	12.0	4.7	<0.001
kcal	2095	2102	452	3032	1173	1985	1972	445	2860	1115	<0.001
Protein											
g	85.9	84.9	18.3	125.1	53.8	85.2	84.4	18.2	126.3	48.7	NS
% food energy	16.7	16.4	2.9	23.1	11.8	17.5	17.2	3.0	24.3	12.5	<0.001
% total energy	15.4	15.1	2.5	21.1	11.2	16.2	15.8	2.6	22.2	11.7	<0.001
Fat											
g	86.5	86.6	24.4	135.3	37.8	80.6	78.2	24.0	134.1	39.2	<.001
% food energy	37.0	36.7	5.7	48.2	25.2	36.3	36.1	5.6	47.4	25.2	0.046
% total energy	34.2	34.6	5.7	45.0	22.4	33.7	33.6	5.7	43.9	23.7	NS
Saturated fatty acids											
g	34.2	33.5	11.9	58.7	13.8	30.1	28.6	10.6	52.4	12.3	<.001
% food energy	14.6	14.3	3.5	21.7	8.3	13.5	13.4	3.1	20.6	8.0	<.001
% total energy	13.5	13.2	3.4	20.3	7.4	12.6	12.5	3.0	19.1	7.3	<.001
Cis mono-unsaturated fatty acids											
g	27.7	27.6	7.9	44.2	12.5	27.5	26.6	8.8	46.5	12.7	NS
% food energy	11.9	11.8	2.1	15.7	7.6	12.4	12.3	2.4	17.7	8.2	<.001
% total energy	11.0	11.0	2.0	14.9	6.9	11.5	11.3	2.3	16.4	7.5	<.001
Cis n-3 poly-unsaturated fatty acids											
g	1.8	1.6	0.8	4.2	0.7	2.2	2.0	0.9	4.7	0.9	<.001
% food energy	0.8	0.7	0.3	1.7	0.4	1.0	0.9	0.4	1.9	0.5	<.001
% total energy	0.7	0.7	0.3	1.5	0.4	0.9	0.8	0.3	1.8	0.5	<.001

	Men 1999 (n=594)					Men 2006-2011 (n=594)					<i>P</i>
	Mean	Median	SD	Upper 2.5 %ile	Lower 2.5 %ile	Mean	Median	SD	Upper 2.5 %ile	Lower 2.5 %ile	
Cis n-6 poly-unsaturated fatty acids											
g	12.5	11.7	5.1	24.5	4.9	11.8	11.0	4.6	21.5	5.3	0.010
% food energy	5.4	5.0	1.8	9.4	2.7	5.3	5.1	1.5	8.8	2.9	NS
% total energy	5.0	4.6	1.8	9.1	2.4	4.9	4.7	1.5	8.2	2.6	NS
Trans fatty acids											
g	3.3	3.1	1.6	7.2	0.9	2.5	2.3	1.2	5.3	0.8	<.001
% food energy	1.4	1.4	0.5	2.6	0.5	1.1	1.1	0.4	2.1	0.5	<.001
% total energy	1.3	1.3	0.5	2.6	0.4	1.1	1.0	0.4	2.1	0.4	<.001
Carbohydrate											
g	259.4	258.1	66.4	394.8	129.9	244.9	244.1	64.6	379.3	117.9	<0.001
% food energy	46.4	46.8	6.1	58.2	34.0	46.2	46.2	5.8	56.3	34.1	NS
% total energy	43.1	43.4	7.3	57.0	26.9	43.1	43.1	6.9	55.5	28.2	NS
Starch											
g	144.4	143.2	37.1	228.8	71.0	133.5	132.7	34.8	207.5	66.6	<0.001
% food energy	26.0	26.0	4.5	35.8	16.7	25.4	25.3	4.6	34.2	16.3	0.036
% total energy	24.2	24.0	4.9	34.1	14.2	23.7	23.6	4.9	33.2	13.7	NS
Sugars											
g	112.8	107.8	44.2	214.6	37.2	109.5	106.7	41.4	197.9	32.6	NS
% food energy	20.0	19.8	6.2	34.0	8.3	20.4	20.5	5.4	31.1	9.5	NS
% total energy	18.6	18.4	5.9	31.9	7.7	19.0	19.2	5.4	29.7	8.3	NS
Alcohol											
g	24.5	17.2	27.7	107.4	0.0	22.5	16.7	23.0	84.9	0.0	NS
% total energy	7.4	5.6	7.5	27.4	0.0	7.2	5.4	7.0	24.9	0.0	NS
Non starch polysaccharide											
g	15.4	14.5	5.0	26.9	7.0	16.8	16.3	5.7	30.3	7.2	<0.001

Table 4.5.7b Average daily intake of energy and macronutrients from food sources of women aged 53 years (1999) and 60-64 years (2006-2011)

	Women 1999 (n=724)					Women 2006-2011 (n=724)					P
	Mean	Median	SD	Upper 2.5 %ile	Lower 2.5 %ile	Mean	Median	SD	Upper 2.5 %ile	Lower 2.5 %ile	
Total energy											
MJ	7.5	7.4	1.5	10.5	5.0	7.3	7.3	1.4	10.2	4.7	0.023
kcal	1776	1758	363	2508	1175	1734	1722	342	2425	1123	0
Food energy											
MJ	7.2	7.1	1.5	10.3	4.6	7.0	6.9	1.4	9.8	4.4	0.005
kcal	1713	1690	360	2455	1095	1662	1642	333	2327	1045	0
Protein											
g	71.3	71.2	14.4	102.5	45.2	71.8	70.9	14.1	102.5	48.8	NS
% food energy	16.9	16.7	3.0	23.2	12.0	17.6	17.3	3.1	24.4	12.2	<0.001
% total energy	16.3	16.1	2.8	22.7	11.6	16.8	16.5	3.0	23.7	11.7	<0.001
Fat											
g	68.1	66.7	21.4	111.4	32.2	66.3	64.7	19.5	108.2	31.5	NS
% food energy	35.3	35.7	6.4	47.2	22.6	35.6	35.6	6.2	47.9	22.9	NS
% total energy	34.0	34.3	6.1	45.5	22.0	34.1	34.3	6.1	45.9	21.9	NS
Saturated fatty acids											
g	26.5	25.2	9.8	48.4	10.6	25.0	23.7	9.1	44.7	9.8	0.004
% food energy	13.7	13.5	3.5	21.1	7.3	13.4	13.4	3.5	20.4	6.8	NS
% total energy	13.2	12.9	3.4	20.6	7.1	12.8	12.9	3.4	20.2	6.5	0.038
Cis mono-unsaturated fatty acids											
g	21.6	20.9	7.1	37.0	9.6	22.0	21.3	6.9	36.7	9.7	NS
% food energy	11.2	11.2	2.4	15.8	6.6	11.8	11.8	2.5	16.9	7.0	<0.001
% total energy	10.8	10.8	2.3	15.0	6.4	11.3	11.4	2.3	16.2	6.8	<0.001
Cis n-3 poly-unsaturated fatty acids											
g	1.4	1.4	0.6	2.9	0.6	1.8	1.6	0.8	3.9	0.9	<0.001
% food energy	0.8	0.7	0.3	1.5	0.4	1.0	0.9	0.4	1.9	0.5	<0.001
% total energy	0.7	0.7	0.3	1.4	0.4	0.9	0.9	0.3	1.8	0.5	0.000

	Women 1999 (n=724)					Women 2006-2011 (n=724)					P
	Mean	Median	SD	Upper 2.5 %ile	Lower 2.5 %ile	Mean	Median	SD	Upper 2.5 %ile	Lower 2.5 %ile	
Cis n-6 poly-unsaturated fatty acids											
g	10.0	9.3	4.4	21.2	4.1	10.0	9.3	3.9	19.6	4.3	NS
% food energy	5.2	4.9	1.8	9.4	2.6	5.4	5.1	1.7	9.3	2.9	NS
% total energy	5.0	4.7	1.8	9.0	2.5	5.2	4.9	1.7	9.2	2.7	NS
Trans fatty acids											
g	2.5	2.4	1.2	5.2	0.7	2.0	1.9	0.9	4.1	0.5	<0.001
% food energy	1.3	1.3	0.5	2.4	0.5	1.1	1.1	0.4	2.0	0.3	<0.001
% total energy	1.3	1.2	0.5	2.4	0.5	1.0	1.0	0.4	2.0	0.3	0.000
Carbohydrate											
g	217.4	213.9	49.9	323.1	130.3	207.4	204.0	49.6	311.9	122.4	<0.001
% food energy	47.7	47.8	6.1	59.7	35.9	46.8	46.9	6.0	57.9	35.2	0.005
% total energy	46.1	46.3	6.7	58.5	32.7	45.0	45.2	6.6	56.9	31.1	0.002
Starch											
g	113.7	112.9	30.1	173.9	60.3	109.1	107.3	30.7	164.6	60.5	0.004
% food energy	25.0	25.0	4.7	33.6	14.6	24.7	24.6	4.9	33.8	15.5	NS
% total energy	24.1	24.2	4.8	33.0	13.5	23.8	23.6	5.0	33.6	14.7	NS
Sugars											
g	101.8	98.3	32.9	174.6	45.3	96.8	94.4	31.2	165.9	44.6	0.003
% food energy	22.4	22.1	5.9	34.7	12.0	21.8	21.6	5.3	32.3	11.7	0.046
% total energy	21.6	21.4	6.0	34.1	11.3	20.9	20.8	5.3	31.6	11.0	0.020
Alcohol											
g	9.5	6.0	10.9	37.1	0.0	10.7	6.2	13.5	43.2	0.0	NS
% total energy	3.8	2.4	4.4	14.2	0.0	4.2	2.4	5.1	16.6	0.0	NS
Non starch polysaccharide											
g	14.5	14.0	4.3	23.8	7.9	15.5	15.1	4.6	26.2	7.9	<0.001

Table 4.5.7c Average daily intake of energy and macronutrients from food sources of all cohort members (men and women) aged 53 years (1999) and 60-64 years (2006-2011)

	All 1999 (n=1318)					All 2006-2011 (n=1318)					<i>P</i>
	Mean	Median	SD	Upper 2.5 %ile	Lower 2.5 %ile	Mean	Median	SD	Upper 2.5 %ile	Lower 2.5 %ile	
Total energy											
MJ	8.4	8.3	2.0	12.6	5.0	8.1	7.9	1.9	12.1	4.8	<0.001
kcal	1996	1963	484	3004	1189	1916	1875	451	2883	1131	<0.001
Food energy											
MJ	7.9	7.8	1.9	11.9	4.7	7.6	7.4	1.8	11.2	4.4	<0.001
kcal	1885	1854	446	2825	1123	1807	1766	419	2677	1045	<0.001
Protein											
g	77.9	76.1	17.8	119.6	47.1	77.9	75.9	17.4	114.3	48.7	NS
% food energy	16.8	16.6	2.9	23.2	11.9	17.5	17.3	3.0	24.4	12.4	<0.001
% total energy	15.9	15.6	2.7	21.6	11.3	16.5	16.2	2.8	23.2	11.7	<0.001
Fat											
g	76.4	74.4	24.5	128.2	34.2	72.7	71.1	22.8	123.7	33.5	<0.001
% food energy	36.1	36.3	6.2	47.5	23.5	35.9	35.9	6.0	47.6	23.5	NS
% total energy	34.1	34.4	5.9	45.3	22.0	33.9	33.9	5.9	45.2	22.4	NS
Saturated fatty acids											
g	30.0	28.6	11.5	55.9	11.4	27.3	26.1	10.1	50.4	10.6	<0.001
% food energy	14.1	13.8	3.5	21.4	7.5	13.5	13.4	3.3	20.5	7.2	<0.001
% total energy	13.3	13.1	3.4	20.5	7.2	12.7	12.7	3.2	19.8	6.8	<0.001
Cis mono-unsaturated fatty acids											
g	24.3	23.6	8.1	41.6	10.3	24.5	23.6	8.3	43.6	10.2	NS
% food energy	11.5	11.5	2.3	15.8	6.9	12.1	12.0	2.5	17.5	7.4	<0.001
% total energy	10.9	10.9	2.1	15.0	6.5	11.4	11.4	2.3	16.3	7.1	<0.001
Cis n-3 poly-unsaturated fatty acids											
g	1.6	1.5	0.7	3.5	0.6	2.0	1.8	0.9	4.2	0.9	<0.001
% food energy	0.8	0.7	0.3	1.6	0.4	1.0	0.9	0.4	1.9	0.5	<0.001
% total energy	0.7	0.7	0.3	1.5	0.4	0.9	0.9	0.3	1.8	0.5	<0.001

	All 1999 (n=1318)					All 2006-2011 (n=1318)					P
	Mean	Median	SD	Upper 2.5 %ile	Lower 2.5 %ile	Mean	Median	SD	Upper 2.5 %ile	Lower 2.5 %ile	
Cis n-6 poly-unsaturated fatty acids											
g	11.1	10.3	4.9	23.3	4.2	10.8	10.0	4.3	20.6	4.5	0.04
% food energy	5.3	4.9	1.8	9.4	2.6	5.3	5.1	1.6	9.2	2.9	NS
% total energy	5.0	4.7	1.8	9.1	2.4	5.1	4.8	1.6	8.7	2.7	NS
Trans fatty acids											
g	2.9	2.6	1.4	6.2	0.8	2.2	2.1	1.1	4.8	0.6	<0.001
% food energy	1.4	1.3	0.5	2.5	0.5	1.1	1.1	0.4	2.0	0.4	<0.001
% total energy	1.3	1.2	0.5	2.4	0.4	1.0	1.0	0.4	2.0	0.3	<0.001
Carbohydrate											
g	236.3	231.3	61.6	374.7	129.9	224.3	218.4	59.8	357.5	119.3	<0.001
% food energy	47.1	47.3	6.1	59.4	34.8	46.6	46.8	5.9	57.2	35.1	0.016
% total energy	44.8	45.1	7.1	58.1	29.3	44.1	44.3	6.8	56.4	30.2	0.02
Starch											
g	127.6	125.4	36.8	209.5	62.0	120.1	117.6	34.8	188.6	62.8	<0.001
% food energy	25.4	25.5	4.7	34.9	15.3	25.0	25.0	4.8	34.2	15.8	0.037
% total energy	24.1	24.1	4.8	33.7	13.6	23.7	23.6	5.0	33.3	14.0	0.04
Sugars											
g	106.7	103.1	38.8	196.2	41.8	102.5	99.2	36.6	182.7	38.5	0.004
% food energy	21.3	20.9	6.2	34.2	10.4	21.2	21.1	5.4	31.8	10.6	NS
% total energy	20.3	19.9	6.2	33.6	9.0	20.1	19.9	5.4	31.2	9.8	NS
Alcohol											
g	16.3	9.7	21.6	70.5	0.0	16.0	9.6	19.3	66.4	0.0	NS
% total energy	5.4	3.5	6.2	21.8	0.0	5.5	3.7	6.2	22.4	0.0	NS
Non starch polysaccharide											
g	14.9	14.2	4.7	25.8	7.4	16.1	15.5	5.2	27.9	7.6	<0.001

Table 4.5.8a Average daily intake of vitamins and minerals from food sources of men aged 53 years (1999) and 60-64 years (2006-2011)

	Men 1999 (n=594)					Men 2006-2011 (n=594)					P
	Mean	Median	SD	Upper 2.5 %ile	Lower 2.5 %ile	Mean	Median	SD	Upper 2.5 %ile	Lower 2.5 %ile	
Vitamins											
Retinol µg	749	431	1392	4401	144	732	385	1385	5704	131	NS
Carotene equivalent µg	3022	2570	2282	8689	407	2746	2408	1704	6718	476	0.018
Thiamin mg	1.7	1.7	0.5	2.7	0.9	1.8	1.7	2.0	2.9	0.9	0.038
Riboflavin mg	2.1	2.0	0.7	3.9	1.1	2.1	1.9	0.7	3.7	0.9	NS
Niacin mg	23.0	22.2	6.7	39.7	12.5	24.2	23.6	7.5	40.4	12.1	0.003
Pyridoxine mg	2.5	2.4	0.7	4.3	1.4	2.7	2.6	0.9	4.6	1.3	<0.001
Vitamin B12 µg	7.2	5.8	5.8	25.8	2.6	7.3	6.0	4.9	23.3	2.4	NS
Folate µg	324	313	110	529	169	349	339	105	589	169	<0.001
Vitamin C mg	95.8	82.3	56.6	243.0	25.6	106.1	95.1	57.9	252.0	26.6	0.002
Vitamin D µg	4.1	3.7	2.2	9.4	1.1	4.2	3.6	2.7	10.6	1.2	NS
Vitamin E alpha tocopherol equivalent mg	11.7	10.9	5.0	25.0	4.2	10.4	10.0	4.0	20.3	4.2	<0.001
Minerals											
Iron mg	12.8	12.5	3.5	20.9	7.1	13.5	13.3	3.8	22.2	6.8	0.001
Haem iron mg	0.8	0.7	0.6	2.3	0.1	0.8	0.7	0.6	2.3	0.1	NS
Non haem iron mg	12.0	11.7	3.4	19.5	6.6	12.7	12.4	3.6	21.0	6.3	0.001
Calcium mg	1038	1003	307	1789	503	971	953	288	1650	494	<0.001
Magnesium mg	327	322	82	502	187	337	333	93	530	168	0.046
Sodium mg	3062	2990	774	4683	1611	2827	2753	769	4693	1499	<0.001
Potassium mg	3595	3539	833	5501	2127	3647	3651	903	5650	1904	NS
Phosphorus mg	1477	1459	330	2209	902	1481	1463	344	2176	802	NS
Copper mg	1.37	1.20	0.89	3.93	0.63	1.48	1.27	0.96	4.40	0.64	0.031
Zinc mg	10.3	10.1	2.5	15.8	6.2	10.5	10.4	2.7	16.3	5.9	NS

Table 4.5.8b Average daily intake of vitamins and minerals from food sources of women aged 53 years (1999) and 60-64 years (2006-2011)

	Women 1999 (n=724)					Women 2006-2011 (n=724)					P
	Mean	Median	SD	Upper 2.5 %ile	Lower 2.5 %ile	Mean	Median	SD	Upper 2.5 %ile	Lower 2.5 %ile	
Vitamins											
Retinol µg	506	348	751	2923	111	522	329	808	3269	104	NS
Carotene equivalent µg	3246	2617	2541	9239	475	2867	2620	1639	6815	663	0.001
Thiamin mg	1.5	1.4	0.4	2.2	0.8	1.5	1.4	0.7	2.6	0.9	0.009
Riboflavin mg	1.8	1.8	0.6	3.2	0.8	1.8	1.7	0.6	3.1	0.8	NS
Niacin mg	18.6	18.3	5.3	29.6	9.6	19.3	18.9	6.0	32.2	10.5	0.025
Pyridoxine mg	2.0	2.0	0.5	3.1	1.2	2.2	2.2	0.7	3.6	1.2	<0.001
Vitamin B12 µg	5.7	5.0	3.3	15.9	2.0	6.1	5.3	3.6	16.1	2.2	0.031
Folate µg	277	271	77	432	145	298	289	101	479	164	<0.001
Vitamin C mg	107.1	97.0	56.4	234.6	29.9	104.1	94.7	48.4	212.1	29.5	NS
Vitamin D µg	3.3	2.9	1.7	7.7	0.8	3.5	3.1	2.1	8.5	0.8	0.010
Vitamin E alpha tocopherol equivalent mg	9.9	9.4	3.7	18.8	4.0	9.4	8.6	3.5	16.6	4.2	0.006
Minerals											
Iron mg	10.9	10.7	2.8	17.2	6.5	11.7	11.4	3.4	18.8	6.3	<0.001
Haem iron mg	0.6	0.5	0.4	1.4	0.0	0.6	0.5	0.4	1.7	0.0	0.010
Non haem iron mg	10.4	10.2	2.7	16.5	6.1	11.1	10.8	3.3	18.1	5.9	<0.001
Calcium mg	919	894	256	1501	455	877	852	241	1374	460	0.001
Magnesium mg	280	276	67	430	168	292	285	71	441	172	0.001
Sodium mg	2409	2375	633	3687	1353	2238	2171	683	3757	1202	<0.001
Potassium mg	3268	3215	708	4790	2045	3279	3222	715	4789	1989	NS
Phosphorus mg	1268	1246	281	1872	769	1284	1276	267	1851	817	NS
Copper mg	1.12	1.02	0.52	2.86	0.60	1.23	1.11	0.56	2.46	0.66	<0.001
Zinc mg	8.4	8.2	2.0	13.0	4.9	8.7	8.6	2.0	13.4	5.4	0.001

Table 4.5.8c Average daily intake of vitamins and minerals from food sources of all cohort members aged 53 years (1999) and 60-64 years (2006-2011)

	All 1999 (n=1318)					All 2006-2011 (n=1318)					P
	Mean	Median	SD	Upper 2.5 %ile	Lower 2.5 %ile	Mean	Median	SD	Upper 2.5 %ile	Lower 2.5 %ile	
Vitamins											
Retinol µg	615	374	1094	4121	113	617	351	1110	4123	116	NS
Carotene equivalent µg	3145	2591	2430	9005	442	2812	2513	1669	6815	577	<0.001
Thiamin mg	1.6	1.5	0.4	2.5	0.9	1.7	1.5	1.4	2.8	0.9	0.003
Riboflavin mg	2.0	1.9	0.6	3.5	1.0	1.9	1.8	0.7	3.4	0.9	NS
Niacin mg	20.6	19.8	6.4	35.6	10.4	21.5	20.6	7.2	37.7	10.7	<0.001
Pyridoxine mg	2.2	2.2	0.6	3.7	1.2	2.4	2.3	0.8	4.3	1.2	<0.001
Vitamin B12 µg	6.3	5.4	4.7	20.9	2.1	6.6	5.6	4.3	19.1	2.2	NS
Folate µg	298	287	96	498	153	321	312	106	551	164	<0.001
Vitamin C mg	102.0	91.5	56.7	241.8	27.0	105.0	94.7	52.9	237.4	28.3	NS
Vitamin D µg	3.7	3.2	2.0	8.8	0.9	3.8	3.3	2.4	10.0	1.0	0.044
Vitamin E alpha tocopherol equivalent mg	10.7	10.0	4.4	21.7	4.1	9.8	9.2	3.8	18.7	4.2	<0.001
Minerals											
Iron mg	11.8	11.4	3.3	19.8	6.7	12.5	12.0	3.7	20.8	6.4	<0.001
Haem iron mg	0.7	0.6	0.5	1.9	0.1	0.7	0.6	0.5	2.0	0.1	NS
Non haem iron mg	11.1	10.7	3.1	18.3	6.2	11.8	11.3	3.5	19.4	5.9	<0.001
Calcium mg	973	941	286	1605	477	919	904	267	1505	476	<0.001
Magnesium mg	301	294	77	482	173	312	305	85	508	171	<0.001
Sodium mg	2703	2656	772	4424	1433	2503	2404	780	4242	1313	<0.001
Potassium mg	3415	3352	783	5233	2067	3445	3385	826	5286	1956	NS
Phosphorus mg	1362	1328	321	2090	793	1373	1341	320	2037	810	NS
Copper mg	1.23	1.09	0.72	3.48	0.61	1.34	1.18	0.78	3.27	0.64	<0.001
Zinc mg	9.3	9.0	2.5	14.9	5.2	9.5	9.2	2.5	15.2	5.5	0.008

Table 4.5.9a Average intake of vitamins and minerals from all sources (foods and supplements) of men aged 53 years (1999) and 60-64 years (2006-2011) as percentage of Reference Nutrient Intake (RNI) and percent of cohort members below Lower Reference Nutrient Intake (LRNI)

	Men 1999 (n=594)				Men 2006-2011 (n=594)			
	Mean %RNI	Median	SD	%below LRNI	Mean %RNI	Median	SD	%below LRNI
Vitamins								
Vitamin A	179	130	206	1.9	169	116	202	0.5
Thiamin	186	185	51	0.2	205	186	217	0.0
Riboflavin	162	156	51	0.5	159	149	54	1.0
Niacin	144	139	42	0.2	151	148	47	0.0
Pyridoxine	179	171	52	0.0	194	185	64	0.0
Vitamin B12	478	390	385	0.2	485	398	325	0.0
Folate	162	156	55	0.2	174	169	52	0.3
Vitamin C	239	206	141	0.3	265	238	145	0.0
Minerals								
Iron	147	144	40	0.0	155	153	44	0.5
Calcium	148	143	44	0.5	139	136	41	0.3
Magnesium	109	107	27	2.9	112	111	31	4.6
Potassium	103	101	24	1.2	104	104	26	2.9
Phosphorus	269	265	60	no LRNI	269	266	63	no LRNI
Copper	114	100	74	no LRNI	124	106	80	no LRNI
Zinc	109	106	27	1.3	110	109	28	1.9

Table 4.5.9b Average intake of vitamins and minerals from all sources (foods and supplements) of women aged 53 years (1999) and 60-64 years (2006-2011) as percentage of Reference Nutrient Intake (RNI) and percent of cohort members below Lower Reference Nutrient Intake (LRNI)

	Women 1999 (n=724)				Women 2006-2011 (n=724)			
	Mean %RNI	Median	SD	%below LRNI	Mean %RNI	Median	SD	%below LRNI
Vitamins								
Vitamin A	174	139	147	0.8	166	132	140	0.7
Thiamin	183	178	47	0.0	193	181	88	0.0
Riboflavin	167	162	53	1.7	164	156	56	2.1
Niacin	155	152	44	0.3	161	157	50	0.0
Pyridoxine	170	167	40	0.0	186	180	58	0.0
Vitamin B12	379	336	223	0.4	405	355	243	0.1
Folate	138	136	39	0.1	149	144	50	0.1
Vitamin C	268	243	141	0.0	260	237	121	0.0
Minerals								
Iron	125	123	32	0.4	135	131	39	0.0
Calcium	131	128	37	1.0	125	122	34	1.4
Magnesium	104	102	25	1.0	108	106	26	0.6
Potassium	93	92	20	2.1	94	92	20	2.9
Phosphorus	231	227	51	no LRNI	233	232	49	no LRNI
Copper	93	85	43	no LRNI	102	92	47	no LRNI
Zinc	120	117	29	0.0	125	123	28	0.4

Table 4.5.9c Average daily intake of vitamins and minerals from all sources (foods and supplements) of all cohort members aged 53 years (1999) and 60-64 years (2006-2011) as percentage of Reference Nutrient Intake (RNI) and percent of cohort members below Lower Reference Nutrient Intake (LRNI)

	All 1999 (n=1318)				All 2006-2011 (n=1318)			
	Mean %RNI	Median	SD	%below LRNI	Mean %RNI	Median	SD	%below LRNI
Vitamins								
Vitamin A	176	136	176	1.3	167	127	171	0.6
Thiamin	184	181	49	0.1	198	183	159	0.0
Riboflavin	165	159	52	1.1	162	154	55	1.6
Niacin	150	146	43	0.2	157	152	49	0.0
Pyridoxine	174	169	46	0.0	189	182	61	0.0
Vitamin B12	423	360	311	0.3	441	377	285	0.1
Folate	149	143	48	0.2	161	156	53	0.2
Vitamin C	255	229	142	0.2	263	237	132	0.0
Minerals								
Iron	135	131	37	0.2	144	138	42	0.2
Calcium	139	134	41	0.8	131	129	38	0.9
Magnesium	106	105	26	1.8	110	107	29	2.4
Potassium	98	96	22	1.7	98	97	24	2.9
Phosphorus	248	241	58	no LRNI	250	244	58	no LRNI
Copper	102	91	60	no LRNI	112	98	65	no LRNI
Zinc	115	113	29	0.6	118	116	29	1.1

Appendix

List of appendices

- Appendix A Flyer of collecting dietary information 2006-2011
- Appendix B Glossary of abbreviations
- Appendix C HNR diet diary used in NSHD 2006-11

Appendix A Flyer of collecting dietary information 2006-2011

MRC NATIONAL SURVEY OF HEALTH AND DEVELOPMENT
University College London Medical School
Department of Epidemiology and Public Health
1-19 Torrington Place
London WC1E 6BT



Collecting dietary information in NSHD

As you know, we have collected dietary information from you at various times during the course of the survey. This information has been very useful for us, and along with our colleagues at MRC Human Nutrition Research in Cambridge, we have been able to examine how diets have changed over time and relate what is eaten to other factors in your life and to your health.



We have compared the diets of the cohort in 1950 to diets of 4 year olds in the 1990s.

Prynne CJ, Paul AA, Price GM, Day KC, Hilder WS, Wadsworth ME.
Food and nutrient intake of a national sample of 4-year-old children in 1950: comparison with the 1990s.
Public Health Nutrition 1999 2:537-547

Now more than ever, scientists are recognising the importance of what we eat for our long term health and well-being, and so as the survey continues, it will become even more important to be able to follow your diet into the future. Once again, we are asking you to record everything you eat for 5 days in a diet diary.

Eating is complicated these days. There are many more foods available in the supermarket and for some categories whole aisles are devoted to one type of product. These often vary in composition in ways that can make quite a difference to intake of specific nutrients. Fat spreads are a good example – there are so many different types – made from corn, soya, olive oil – and then some are low fat, others ultralow – and so we need to know as much detail as you can give us.

There have been other changes too. Many people in the UK are now eating “ready meals” several times a week. These also vary in what they contain. To help us analyse the records correctly, we are asking you to collect the labels from products like these. It will give us the detail we need, and save you having to write so much in the diary.

We know that recording all you eat can be tiresome, but the information you provide will be enormously valuable for examining the importance of the diet throughout life.

Thanks for your help!

Appendix B Glossary of abbreviations

AER	Average Energy Requirement
ANOVA	Analysis of variance
DINO	Diet In Nutrients Out
DRV	Dietary Reference Value
EAR	Estimated Average Requirement for energy
HNR	MRC Human Nutrition Research
LHA	MRC Unit of Lifelong Healthy Ageing
LRNI	Lower Reference Nutrient Intake
MOU	Memorandum of Understanding
NSHD	National Survey of Health and Development
RNI	Reference Nutrient Intakes

Appendix C HNR diet diary used in NSHD 2006-11

Day <u>Friday</u>				Date <u>5th February 2010</u>	
Time slot	When	Where	With whom	Food/Drink description & preparation	Portion size
6am to 9am	7-10am	Kitchen	Husband	Pure orange juice - Smooth Tea with Semi-skimmed milk Toast - white - thick sliced Flora light marmalade - orange Fine cut Grandessa (Aldi)	1 Glass 1 Mug A little 2 slices med spread med spread
9am to 12 noon	10.05	Kitchen	alone	Instant Coffee - Nescafe Original Decaff with Semi-skimmed milk	Mug a little

12 noon to 2pm	12.15	Kitchen	Husband	Cheese cobs Cobs - Asda's wholemeal - Small Flora light Cathedral City mature cheddar Pickled onions - Small - Sainsburys Salt Asda's low fat probiotic blackcurrant yogurt Asda's Special fruit score Flora light medium sized Orange Pepsi max	2 Cobs medium spread Thin Sliced 2 Sprinkled 1 1 medium spread 1 1 Glass
2pm to 5pm	3pm	at Neighbours	With Friends	Instant Coffee - Maxwell House Semi-Skimmed Milk	1 Mug a little

Time slot	When	Where	With whom	Food/Drink description & preparation	Portion size
5pm to 8pm	5.50pm	Kitchen	Husband	Steak Pie & Chips with mushy peas & Curry Sauce Steak Pie - Asda's own Chips from Fish n Chips Shop - mushy peas from Fish n Chip Shop Curry Sauce from " " " Salt. Pepsi Max Rhubarb Yogurt lowfat Asda's own Banana Pear - skin on	(8) B (5) B 1 Tablesp. 1 Tablesp. Shaken over; 1 Glass, 1
8pm to 10pm	8.20pm	Lounge	Husband	Pepsi Max	1 Glass
10pm to 6am					

Space to write in recipe or ingredients of made up dishes or take-away food if not already described

Please describe any vitamins, minerals or other food supplements you may have taken

Brand	Name (in full)	Number of pills, capsules, teaspoons
Simply Supplements	Glucosamine Sulphate 500mg	1 tab.
Health Essentials	Evening Primrose Oil 500mg	2 caps.

General Questions about your food / drink last week.

1. Did you follow a special diet? e.g. vegetarian, cholesterol lowering, weight reducing, etc. Yes ☐ please specify No ☒
2. Which type of milk did you use most often last week? Full fat ☐ Dried ☐ Semi-skimmed ☒ Skimmed (fat free) ☐
Soya ☐ type Other ☐ type
3. How much milk did you usually have in coffee / tea? Coffee Tea A lot A lot Average Average Hardly any Hardly any
4. Did you drink decaffeinated coffee / tea last week? Coffee Tea Always Always Sometimes Sometimes Rarely / Never Rarely / Never
5. Which types of fat did you use last week for baking, frying, spreading & salads?
- | | Brand name | Baking | Frying | Spreading | Salads |
|----------------------|--------------------|--------|-------------------------------------|-------------------------------------|--------|
| Butter | | | | | |
| Low fat spread | Flora Omega 3 Plus | | | ☒ | |
| Very low fat spread | | | | | |
| Polyunsaturated | | | | | |
| Margarine | | | | | |
| Other soft margarine | | | | | |
| Hard margarine | | | | | |
| Vegetable oils | Aliso Dry | | ☒ | | |
| White vegetable fat | | | | | |
| Lard | | | | | |
| Dripping | | | | | |
| Other | | | | | |

6. Which type of bread did you eat most often last week? White ☒ Granary ☐ Wholemeal ☐ Softgrain ☐
Brown, Wheatgerm, Hovis ☐ Other ☐ type
7. How thickly did you spread butter, margarine etc. on bread, biscuits? Thick ☐ Medium ☒ Thin ☐ None ☐
8. If you ate meat last week, what did you do with the visible fat? Ate all ☐ Ate most ☐ Ate some ☐ Ate none ☒
9. If you ate poultry last week, did you eat the skin? Yes ☐ No ☒ Sometimes ☐
10. Was salt added to your food during cooking last week? Yes ☐ No ☒ Don't know ☐
- Did you usually add salt to your food at the table? Yes ☒ No ☐
- Did you regularly use a salt substitute (e.g. LoSalt) last week? Yes ☐ brand No ☒
11. Did you eat the skin on fruit?
- | | Apple | Pear | Nectarine | Peach | Other (type) |
|----------------|-------|-------------------------------------|-----------|-------|--------------|
| Skin eaten | | ☒ | | | |
| Skin not eaten | | | | | |
12. Which type of water did you use most often last week? Tap ☐ Filtered ☒ Bottled ☐ brand

Please tell us about anything else about your diet last week that you feel is important.

Thank-you very much for completing this diary. Please return it to the study team as requested.